

DCS 5.5 Core Preprint: A Typed Certification Framework for Cross-Scale Causal Interfaces

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Abstract

Dynamical Causal Structuralism (DCS) 5.5 studies how typed causal structures, lawful local rewrites, and registered dynamics generate multiway histories, conditionally persistent organizations, and cross-scale effective interfaces. We distinguish structural reachability causality, generative history dynamics, and response-intervention causality, and register event quotients, history-response quotients, state abstractions, and intervention implementations as four distinct cross-scale map types; a task contract must register admissible actions, future targets, time horizon, environment, distances, and componentwise error budgets. The strict results of this paper are confined to finite-state, discrete-time, classical Markov models under frozen task contracts: the Finite Exact Interface Quotient Theorem (CPIR-T1) constructs a typed interface package in which event, history, state, and action quotients coexist under six explicit premises; the Persistence Probability Preservation Corollary (CPIR-C1) holds only for quotient-measurable persistence events; and the Finite Approximate Error Theorem (CPIR-T2) provides an explicit finite-horizon accumulated error bound in total variation distance, distinguishing the contractive bound from the linear finite-horizon bound when the Dobrushin coefficient equals one. Conditional causal persistence registers identity, formation rate, survival, recovery, maintenance cost, dynamical closure, and intervention exchange as independently fallible certification axes. All numerical results are executable certificates and mandatory counterexamples (CPIR-X1 through CPIR-X8) within finite models; they do not constitute empirical confirmation in natural systems. The paper does not claim a derivation of four-dimensional Lorentzian spacetime, quantum field theory, the Standard Model, life, intelligence, or civilization, nor a general composition theorem for open interconnected systems. Its contribution is an auditable, type-safe, counterexample-aware certification framework for cross-scale causal interfaces.

Keywords: causal order; graph rewriting; finite-state dynamics; conditional persistence; causal abstraction; intervention consistency; history quotient; causal compression; CPIR; counterexample governance

Claim Status and Citation Policy

The propositions in this paper are labeled as external standard results, conditional DCS results, conjectures, or future work, and any use of the terms “theorem,” “corollary,” “conjecture,” “certificate,” or “empirical support” is consistent with that labeling. Formal citation information follows the post-publication DOI record; correspondence and ORCID remain author-controlled public meta-data.

Declarations

- **Code availability:** The source package, model code, configuration files, and release manifest are frozen for RC6. A public repository URL and archival DOI will be supplied with the public release record.
- **Data availability:** All numerical results are executable certificates generated from the registered model files; complete parameters, seeds, and run commands appear in Appendix D. No external datasets are required to reproduce them.
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1 Introduction

1.1 The Multi-Scale Object Qualification Problem

Many fields of modern science face the same problem: a system is described at the micro level by a large number of states and events, while research, prediction, and control must be carried out at some coarser macro level. The question is: which macro objects are qualified to be treated as “the same object” persisting, being intervened upon, and being predicted? And which macro laws are qualified to be called “the laws of that system”?

Existing answers come from several independent traditions: finite posets and causal sets describe event-level structure, graph rewriting describes lawful local transformations, Markov chains and semigroups describe finite-state dynamics, lumpability and coarse-graining describe macro dynamical closure, causal abstraction describes intervention semantics, and computational mechanics and the information bottleneck describe predictive compression. Each tradition solves one facet, but the facets lack a unified typed contract: events, states, histories, and actions are forced into a single untyped map or “one causal arrow.”

1.2 What Existing Mature Results Each Solve

- Finite posets and Hasse diagrams answer the structural question of whether one event can reach another (Bombelli et al. 1987; Norris 1997);
- Graph rewriting answers the generative question of which transformations are lawful;
- Finite conservative Q matrices and stochastic semigroups answer the dynamical question of how states evolve, in both discrete and continuous time (Norris 1997; Kemeny and Snell 1976; Anderson 1991);
- Strong lumpability answers the question of which partitions yield autonomous macro Markov dynamics (Kemeny and Snell 1976; Buchholz 1994);
- Causal abstraction answers when macro and micro interventions commute (Rubenstein et al. 2017; Beckers and Halpern 2019);
- Computational mechanics, information theory, and the information bottleneck answer what the minimal predictively sufficient state is, and how much information coarse graining discards (Crutchfield and Young 1989; Shalizi and Crutchfield 2001; Tishby et al. 1999; Cover and Thomas 2006);
- Mori–Zwanzig projection methods answer how eliminated degrees of freedom generate memory (Zwanzig 1961; Mori 1965).

These results are mature mathematics; this paper does not claim originality for them.

1.3 The Missing Joint Typed Contract

The tools above each hold in their own domain, but “the same macro interface passing structural, dynamical, response, intervention, memory, and persistence checks simultaneously” lacks a unified form. Concretely: a partition may satisfy state lumpability yet destroy the event poset; a history compression may preserve the predictive distribution yet lose intervention semantics; a macro kernel may be dynamically closed yet be non-measurable for persistence events. The entry point of DCS 5.5 is to register these checks as certification axes of one task contract, typed so as not to be confused with one another, rather than to build a single untyped map that does everything.

1.4 Contributions of This Paper

The core contribution of this paper is an auditable, type-safe, counterexample-aware certification framework for cross-scale causal interfaces:

1. **Type separation:** structural reachability causality, generative history dynamics, and response-intervention causality are treated as distinct duties; event quotients, history-response quotients, state abstractions, and intervention implementations are registered as four distinct cross-scale map types;
2. **Task contracts:** a general task contract and a finite instance task contract are registered separately, containing admissible actions, future targets, time horizon, environment, distances, and componentwise error budgets;
3. **Conditional persistence:** conditional causal persistence registers identity, formation rate, survival, recovery, maintenance cost, dynamical closure, and intervention exchange as independently fallible certification axes, with the survivor conditioning enrichment identity (CP-T0) and the finite four-quadrant counterexample (CP-T1);
4. **Exact finite quotient:** CPIR-T1 proves under six premises that event, history, state, and action quotients coexist within one frozen task contract while preserving their respective types; CPIR-C1 proves that the probability of quotient-measurable persistence events is preserved under the exact quotient;
5. **Finite approximate error bound:** CPIR-T2 gives an explicit finite-horizon accumulated error bound in total variation distance, distinguishing the contractive bound from the linear bound when the Dobrushin coefficient equals one;
6. **Executable counterexample governance:** the eight mandatory counterexamples CPIR-X1–X8 kill, respectively, arbitrary poset quotients, automatic SCC inheritance, arbitrary Markov quotients, uniqueness of multi-implementation effects, automatic task sufficiency from dynamics, automatic multi-step memorylessness from one-step fits, automatic identity preservation by dynamical quotients, and automatic capability generation by compression.

1.5 What This Paper Does Not Claim

This is a preprint focused on the finite mathematical core of DCS 5.5, not the complete treatise, and not DCS 6.0. This paper does not claim: a derivation of four-dimensional Lorentzian space-time, quantum field theory, or the Standard Model from DCS; a composition theorem for open interconnected systems; a unified explanation of life, intelligence, or civilization; or an intelligence

capability lattice or collective agency certification. CPIR-T1/T2 are conditional results in finite, classical, registered task contracts; the new contributions of DCS are mainly typed stitching, certification structure, failure exits, and governance, not the reinvention of posets, lumpability, Markov contraction, or causal abstraction (Kemeny and Snell 1976; Buchholz 1994; Rubenstein et al. 2017; Beckers and Halpern 2019; Andrieux 2012).

At the same time, the framework is designed to connect to the wider research programs that motivate cross-scale questions: free-energy and active-inference accounts of brain function treat perception and action as a single inferential economy (Friston 2010); autopoiesis theory asks what it means for an organization to persist as an identity (Maturana and Varela 1980); and reinforcement learning formalizes agency as the pursuit of future targets under action (Sutton and Barto 2018). DCS 5.5 does not explain any of these programs; it provides a type-safe registration system in which their cross-scale claims can be stated, checked, and failed item by item. This is why the paper insists that events, states, histories, and actions never collapse into one untyped arrow: each research tradition has a legitimate facet, and the certification question is exactly which facets coexist under a frozen contract.

1.6 Paper Structure

Section 2 gives the typed DCS/CPIR theoretical framework; Section 3 gives finite structure, generation, and dynamics foundations; Section 4 discusses conditional persistence, causal abstraction, and selective compression; Section 5 gives the finite exact CPIR interface quotient; Section 6 gives the finite approximate CPIR error bound; Section 7 reports validation analysis and failure exits; Section 8 discusses related work, originality, and limitations; Section 9 summarizes and lists future work. Full proofs, the counterexample library, numerical models, and the proposition ledger are developed in Appendices B, C, D, and E, respectively.

The introduction establishes the problem: cross-scale macro objects lack a unified typed qualification contract. The path of this paper is to make structure, dynamics, response, intervention, memory, and persistence independently certifiable axes, to give an exact quotient and an approximate error bound within finite classical models, and to register explicitly every assertion beyond the finite scope as a conjecture or future work.

2 The Typed DCS/CPIR Theoretical Framework

This section is the conceptual center of the paper. Every map is typed; any diagram must place events, states, histories, and actions in separate lanes, and an untyped ϕ or “one causal arrow” is forbidden from carrying all maps.

Why types, and why now. The intuition behind the typing discipline is close to the oldest lesson of structural realism: the structure that makes a system what it is should not be conflated with the implementation that realizes it (Ladyman 1998). A causal structure can be read off from the geometry of its intervention and response relations (Chvykov and Hoel 2021), and abstraction procedures exist that reason explicitly about lossy macro representations (Xia and Bareinboim 2025). Once these traditions meet, a single untyped map is no longer sufficient: the same symbol would have to mean “can reach” in one sentence, “realizes the dynamics” in the next, and “changes after an action” in the third. DCS 5.5 makes this separation the first-class object of the framework. The reader should think of a typed interface not as a piece of algebra but as a contract with named obligations: structural reachability is one duty, dynamical realization another, response-intervention semantics a third, and no single arrow may discharge two duties at once.

2.1 Three Causal Duties

DCS distinguishes three mutually non-identical causal duties:

$$\boxed{\mathcal{C}_r^{ord} \xrightarrow{\mathcal{D}_r} \mathcal{C}_r^{resp}}$$

The meaning: structure decides “whether one can reach,” dynamics decides “how it is realized,” and response semantics decides “what changes after an action.” This is not two competing ontological roots, nor an identification of the three. Structural reachability causality provides only precedence and reachability constraints; dynamical realization requires generators, kernels, or history weightings; response-intervention causality requires action and target semantics (Pearl 2009; Rubenstein et al. 2017; Beckers and Halpern 2019).

2.2 The Full Registration System

A full registration system at scale r is written as:

$$\mathfrak{C}_r = (E_r, \preceq_r; X_r, \mathcal{H}_r; \mathcal{A}_r, \mathcal{D}_r; \mathcal{O}_r).$$

Here E_r is the event set with event poset $\preceq_r$; X_r is the state space and $\mathcal{H}_r$ the history space; $\mathcal{A}_r$ is the action set and $\mathcal{D}_r$ the dynamics; $\mathcal{O}_r$ is the observable object set. The five dependency graphs (proof, evidence, philosophical argument, bridging, version migration) are registered separately and never merged into one graph.

2.3 Four Distinct Cross-Scale Maps

Cross-scale maps are not a single coarse-graining but four type-distinct maps:

$$q_r^E : E_r \rightarrow \bar{E}_r, \quad \Phi_r^H : \mathcal{H}_r \rightarrow Z_r,$$

$$\tau_r^X : X_r \rightarrow \bar{X}_r, \quad \Gamma_r^A : \bar{\mathcal{A}}_r \rightrightarrows \mathcal{A}_r.$$

Map	Admissible input	Admissible output	Primary admission
q_r^E	events and event relations	event quotient	poset contract and structure preservation
Φ_r^H	full or truncated histories	interface/memory state	task response sufficiency
τ_r^X	micro states/state distributions	macro states/pushed-forward distributions	dynamical closure and representation error
Γ_r^A	macro actions	sets of micro implementations	implementability and implementation invariance

Rewriting all four as a single untyped **phi** is forbidden. Γ_r^A is a set-valued implementation correspondence; it must not be silently converted into a single-valued function.

2.4 Task Contracts

The general task contract is registered at scale r :

$$\Theta_r = (\mathcal{A}_r^\Theta, F_r, T_r, \mathcal{E}_r, d_r, \epsilon_r).$$

Here $\mathcal{A}_r^\Theta$ is the admissible action set, F_r the future target, T_r the time horizon, $\mathcal{E}_r$ the environment, d_r the distance, and ϵ_r the componentwise error budget. In the 09E finite model the finite instance is used:

$$\Theta = (\Pi, F, T, \mathcal{E}, d, \delta).$$

The two are a general specification and a finite instance; their similar symbols must not lead to merging fields.

2.5 The D0/D1/D2 Closure Ladder

Closure is registered by grade:

- **D0-exact**: the macro kernel is exactly autonomous on the registered partition;
- **D1-approximate**: the macro kernel is approximately autonomous within the registered distance and error budget;
- **D2-memory-augmented**: the macro state requires augmented memory to recover task-relative closure.

D2 memory augmentation is written as:

$$\tilde{Z}_t = (Z_t, M_t),$$

with memory residual:

$$\varepsilon_{\text{mem}} = I(F_{t:T}; H_t^- \mid Z_t, M_t, A_{t:T}).$$

When the first-order Markov assumption fails, one registers memory under D2 rather than hiding the memory or claiming memoryless closure.

2.6 Three Profiles

- **CPIR-C**: full CPIR certification, requiring structure, dynamics, response, memory, intervention, and capability to pass item by item;
- **CPIR-Q**: quantitative scoring only, without full certification;
- **PRE-CPIR**: pre-certification research stage, with only some axes checked.

Different profiles may use only a type-safe subset of the four maps: q^E is enabled only when changing event granularity; Γ^A is mandatory only when claiming high-level intervention qualification.

2.7 Three Scorecards and the Independent Persistence Passport

Error vector:

$$\mathbf{E}_r = (\varepsilon_{\text{ord}}, \varepsilon_{\text{dyn}}, \varepsilon_{\text{resp}}, \varepsilon_{\text{impl}}, \varepsilon_{\text{int}}, \varepsilon_{\text{mem}}),$$

compression gain vector:

$$\mathbf{Q}_r = (R_{\text{state}}, R_{\text{history}}, R_{\text{parameter}}, R_{\text{compute}}),$$

capability result vector:

$$\mathbf{K}_r = (\mathcal{R}_{\text{reliable}}, C_{\text{ctrl}}, T_{\text{pred}}, R_{\text{robust}}, G_{\text{env}}).$$

The three tables use itemwise thresholds, confidence intervals, and Pareto relations; any aggregate score must attach weight sources, sensitivity analysis, and incomparability notes, and does not serve as CPIR base admission. They are compared with the C5 persistence passport by constraint/Pareto relations and must not be synthesized into a “universe evolution total score.” ε_{dyn} , $\varepsilon_{\text{resp}}$, $\varepsilon_{\text{impl}}$, ε_{int} , ε_{mem} , and $\varepsilon_{\text{pers}}$ are listed separately; one must not substitute for another.

2.8 Interface Certification and Causal Role Promotion

Given a task contract Θ_r , interface certification is written as:

$$I_r = \text{Cert}_{\Theta_r}(\mathfrak{C}_r; q_r^E, \Phi_r^H, \tau_r^X, \Gamma_r^A).$$

Then the typed embedding:

$$\iota_r : I_r \hookrightarrow \mathfrak{C}_{r+1}$$

gives it state identity and action identity at the next layer. This “causal role promotion” must preserve lower-layer implementation fibers, resource and environment conditions, effective task domain, error certificates, and failure exits. Embedding means that the interface acquires a referenceable, composable, intervenable role in the higher-layer model; it does not mean that the lower-layer implementation class and the higher-layer cause class are set-theoretically identical. If the SCC is repaired, the state partition is reselected, memory variables are changed, or the intervention implementation set is replaced, the candidate interface has changed and all related poset, dynamics, response, intervention, closure, and persistence certifications must be rerun; old certificates serve only as historical records.

2.9 Exact/Approximate Dual Track

CPIR certification runs on two parallel tracks:

- **Exact track:** under finite, exact, classical, frozen task contracts, quotient maps commute exactly (CPIR-T1/C1 in Section 5);
- **Approximate track:** under finite, discrete-time, classical Markov models and total variation distance, error accumulates recursively over a finite horizon (CPIR-T2 in Section 6).

The two tracks share the same type system and task contract format; exactness is the limiting case of approximation, and approximation does not sacrifice type safety.

This section establishes nine components of the typed framework: three causal duties, the full registration system, four cross-scale map types, task contracts, the D0/D1/D2 closure ladder, three profiles, three scorecards with the independent persistence passport, interface certification with causal role promotion, and the exact/approximate dual track. All theorems in subsequent sections are stated within this framework.

3 Finite Structure, Generation, and Dynamics Foundations

This section keeps only the minimal foundations supporting later theorems. Standard proofs are given as proof sketches in the main text with references to Appendix B; standard mathematical results must not be packaged as original DCS theorems.

Why finite models. The entire strict core of this paper is stated on finite objects, and the choice is deliberate rather than convenient. A finite state space and a finite horizon make every certification axis checkable in principle: one can enumerate counterexamples, exhaust partitions, and freeze a task contract without approximation. Finite models are also where the classical tools this paper reuses were built — continuous-time Markov chains are most tractable in their finite, conservative form (Anderson 1991), and the information-theoretic quantities used below are only unproblematic when the state space is finite (Cover and Thomas 2006). Finally, finiteness is not an exotic restriction for the natural systems this framework aims to serve: the collective statistics of neural activity — neuronal avalanches — are typically measured on finite spike trains with precisely this structure (Beggs and Plenz 2003), and cortical dynamics exhibit a hierarchy of intrinsic timescales that a finite task contract is designed to register rather than erase (Murray et al. 2014). Nothing in this section is original mathematics; the point is to state the finite foundations cleanly enough that the later theorems inherit them without hidden assumptions.

3.1 Finite Posets and Strict Relations

Definition (strict causal poset, C1-A1). A relation $\prec$ on a set V is a strict poset if and only if for all $x, y, z \in V$:

$$\neg(x \prec x), \quad (x \prec y \wedge y \prec z) \Rightarrow x \prec z.$$

Proposition (Hasse diagrams are acyclic, C1-P1). The Hasse diagram of a finite strict poset is a directed acyclic graph. This follows from the transitivity contradiction of cover relations.

Theorem (every finite poset has a linear extension, C1-T1, inherited external standard result). Every nonempty finite poset has a minimal element and admits a linear extension. Proof sketch: finiteness forbids infinite descending chains; take minimal elements by induction and place them first; full proof in Appendix B.1.

Theorem (transitive reduction of a finite DAG is unique, C1-T2, inherited external standard result). The transitive reduction of a finite DAG is unique and equals the Hasse diagram of its reachability poset. Proof sketch: cover edges must be kept, otherwise the transitive closure contracts; full proof in Appendix B.2.

Conditions and boundaries. These conclusions hold only for finite strict posets. Local finiteness does not imply finite past (Bombelli et al. 1987); an arbitrary poset does not automatically have a manifold embedding or spacetime meaning. Implementation obligation: “zero in-degree” in programs must be computed on the Hasse diagram or on a DAG consistent with the transitive closure; inputs with cycles must be rejected before sorting.

3.2 Hasse Diagrams and Transitive Reduction

The Hasse diagram is the minimal redundant-free representation of the transitive closure; cover edges must not be stored together with the transitive closure. Inserting intermediate points changes cover relations (counterexample D-C2-01, Appendix C.2). Structural identity is determined by the poset itself, not by the geometric layout of the diagram.

3.3 Lawful Configurations and Local Rewrites

Definition (lawful rule sequence, C2-P1). Let $r_k : D_k \subseteq \Omega \rightarrow \Omega$ all be lawful, i.e.,

$$r_k(D_k \cap \Omega) \subseteq \Omega.$$

Proposition (lawful finite rule sequences preserve closure of the configuration space, C2-P1). For any finite executable lawful rule sequence $r_n \circ \dots \circ r_1$, the output remains in the lawful configuration space Ω . Proof sketch: induction on sequence length; full proof in Appendix B.3.

Conditions and boundaries. This proposition guarantees neither that an executable rule always exists, nor that rewriting terminates, nor that continuous-time processes are non-explosive. Configuration transition edges (state jumps) are not event poset edges: state jumps are objects of the dynamics layer, event poset edges are objects of the structure layer, and the two must not be conflated (Bombelli et al. 1987; Rideout and Sorkin 2000).

3.4 Finite Conservative Q Matrices and Stochastic Semigroups

Definition. A finite conservative Q matrix $Q = (q_{ij})$ has nonnegative off-diagonal entries and zero row sums, with departure rates $q_i = -q_{ii}$.

Theorem (a finite conservative Q generates a unique stochastic semigroup, C3-T1, inherited external standard result). A finite conservative Q matrix generates a unique transition semigroup $P_t = e^{tQ}$. Proof sketch: uniformize $K = I + Q/\lambda$ into a stochastic matrix; positivity follows from a Poisson mixture decomposition; full proof in Appendix B.4.

Theorem (a finite irreducible CTMC has a unique stationary distribution and converges, C3-T2, inherited external standard result). A finite irreducible CTMC has a unique strictly positive stationary distribution, and every initial distribution converges to it as $t \rightarrow \infty$. Proof sketch: closed communicating class plus the Perron–Frobenius theorem; full proof in Appendix B.5.

Theorem (relative entropy contraction, C3-T3, inherited external standard result). For any Markov kernel P and probability measures μ, ν :

$$D(\mu P \| \nu P) \leq D(\mu \| \nu).$$

In particular, the KL divergence to the stationary distribution is nonincreasing under the Markov semigroup. This is the Markov data-processing property, not “a universal increase of Shannon entropy,” nor a law of thermodynamic entropy production. Full proof in Appendix B.6.

Conditions and boundaries. These conclusions are confined to finite states, conservative Q , and registered time conventions. Growing systems generally have no fixed full-state stationary distribution (counterexample D-C3-02); continuous-weight models cannot be treated as finite-state (failure mode 32.1).

3.5 Boundaries of Stationarity and Convergence

Existence of a stationary distribution does not imply local organization (counterexample D-C5-01); identical state graphs with different rates can yield metastable lifetimes differing by arbitrary orders of magnitude (counterexample D-C5-02); a global spectral gap does not guarantee that a specified partition is blockable (counterexample D-C4-03). Dynamical indicators and purely topological spectra must be used separately (Norris 1997; Bovier and Hollander 2015).

3.6 Strong Lumpability

Theorem (strong lumpability induces macro Markov dynamics, C4-T1, inherited external standard result). Let a finite state space be partitioned by $C : \Omega \rightarrow \bar{\Omega}$. If any two states $i, j \in B_\alpha$ in the same source block have equal total rates into every target block B_β , then there exists a unique conservative macro generator $\bar{Q}$ such that the macro semigroup $\bar{P}_t$ satisfies $C_\# P_t^* = \bar{P}_t^* C_\#$. Full proof in Appendix B.7.

Conditions and boundaries. The theorem requires a fixed partition and the strong lumpability condition; when coarse-graining generates memory, block sojourn times are non-exponential (counterexample D-C4-01), and Markov closure cannot be claimed automatically.

3.7 Entering D2 When First-Order Markov Fails

When eliminated degrees of freedom produce persistent history dependence, the correct procedure is to enter D2 memory augmentation registration (Section 2.5), registering the memory residual ε_{mem} explicitly rather than hiding the memory or claiming memoryless closure. Macro autonomy holds only within the registered time window (Zwanzig 1961; Mori 1965); the Zwanzig nonlinear transport derivation is narrowed according to the 1972 correction (Zwanzig 1961; Zwanzig et al. 1972).

All results in this section are inherited finite-model tools; DCS types and implements them, and does not claim them as original theorems.

4 Conditional Persistence, Causal Abstraction, and Selective Compression

4.1 Persistence Is Not Stasis

The intuition behind the section. Across the life sciences, the hardest object to certify is often not a quantity but an identity: what does it mean for a living organization to “keep being itself” while its material composition turns over continuously? Chemistry gave one of the earliest answers in the form of self-organizing autocatalytic cycles — the hypercycle — in which a population of replicators maintains itself through mutual catalysis rather than stasis (Eigen 1971; Eigen and Schuster 1979). Autopoiesis theory sharpened the same idea into a criterion: an organization persists when its components are continuously produced by the very network they constitute (Maturana and Varela 1980). Evolutionary theory then added the quantitative handle of selection and covariance, which tracks how a population-level quantity changes under differential persistence (Price 1970), and the theory of major transitions showed that persistence can migrate to new carriers — from molecules to cells to societies — each time with new maintenance costs (Maynard Smith and Szathmary 1995). DCS 5.5 takes this whole lineage and turns it into a certification problem: persistence is not “the object being unchanged”; it is a conditional claim about identity, formation, survival, recovery, and cost under a registered contract, and each component can fail independently.

Definition (conditional contract for causal persistence states, CP-D1). Persistence is not “the object being unchanged”; it is the continued existence, in a statistical population sense, of a pattern under a given identity relation, scale, environment, perturbation, and time window, accompanied by resistance, recovery, or reconstruction. The persistence passport `PersistenceClaim` registers at least object, identity, model, scale, time window, environment, perturbation, indicator, cost, null model, evidence, and failure conditions (specification in Appendix E.4).

4.2 Formation, Retention, Recovery, Rebuilding, Cost, and Failure Are Separately Registered

Cross-sectional abundance is jointly determined by formation rate and lifetime. **Theorem (survivor conditioning enrichment identity, CP-T0, conditional DCS result).** Let Z be a pattern feature measured at formation, and $S_T = \{\tau_{Ac} > T\}$ the event of surviving to time T under a pre-registered identity contract. Then for any measurable set B :

$$\Pr(Z \in B \mid S_T) = \frac{\Pr(S_T \mid Z \in B) \Pr(Z \in B)}{\Pr(S_T)}.$$

Conditions and boundaries. This identity only states composition change under conditional sampling. It neither identifies a causal effect of Z on survival, nor implies that the system actively maximizes Z , lifetime, or any scalar “persistence potential.” Common causes, formation-rate differences, measurement error, and selection bias can all produce the same observational result; causal interpretation requires separately registered interventions, identification assumptions, or mechanistic evidence. Full proof in Appendix B.8.

Theorem (finite models realize the persistence–EI four quadrants, CP-T1, conditional DCS result). There exists a class of finite discrete-time Markov models whose four deterministic kernels realize, respectively, “high persistence–high EI,” “high persistence–low EI,” “low persistence–high EI,” and “low persistence–low EI,” where one-step effective information is defined as $EI_1 = I(I; Y)$ (base-2 logarithm) and the persistence probability is $S_A(T) = \Pr(\tau_{Ac} > T)$. The theorem negates the logical identification of EI with persistence; it does not negate statistical association in specific empirical model families. Full construction and proof in Appendix B.9.

4.3 Observational Commutation, Dynamical Closure, Intervention Commutation, and Counterfactual Commutation Are Not the Same

Multi-scale causal abstraction registers four kinds of commutation:

- **Observational commutation:** $C_{\#}P_M = P_{\bar{M}}$ (passive predictive agreement);
- **Dynamical closure:** the macro projection is approximately autonomous within the registered time window;
- **Intervention commutation:** $(C^{abs})_{\#}P_M^{do(i)} = P_{\bar{M}}^{do(\vartheta(i))}$ (macro–micro intervention consistency);
- **Counterfactual commutation:** $C_{\#}P_M^{do(i|\text{counterfactual condition})}$ agrees with the macro counterfactual.

Passive dynamical closure does not substitute for causal abstraction; macro–micro state maps and intervention maps must approximately commute under the withheld interventions (Pearl 2009; Rubenstein et al. 2017; Beckers and Halpern 2019; Beckers et al. 2020).

Theorem (composition preservation of compatible exact causal abstractions, C4I-T1, conditional DCS result). If two levels commute exactly and states, interventions, and initial conditions are compatible, then the composite map preserves exact commutation. **Boundary:** this is a two-level, exact, domain-compatible conditional result; general composition preservation for open systems is P2B future work, not a conclusion of this paper.

4.4 An Effective Macro Interface Must Pass Five Checks Simultaneously

Definition (two-stage minimally sufficient persistence causal interface, SCC-D2). An effective macro interface must simultaneously pass: response separation (distinguishable class-level responses), genuine compression, dynamical closure, memory, intervention, and persistence hard admissions, and only among admissible partitions compare Pareto (Buchholz 1994; Andrieux 2012).

Model K and the cross-scale interface prototype (Appendix D.4) show that a partition may simultaneously satisfy response, closure, memory, intervention, and persistence checks (exact special case), or fail them simultaneously under within-block heterogeneity (counterexample CSR-P2); Pareto membership itself does not constitute an effective-layer certification.

4.5 Information Compression, Predictive Compression, and Causal Compression Are Separate Accounts

Information compression (statistical redundancy reduction), predictive compression (predictively sufficient states (Crutchfield and Young 1989; Shalizi and Crutchfield 2001; Tishby et al. 1999)), and causal compression (intervention-semantics preservation (Rubenstein et al. 2017; Beckers and Halpern 2019)) are three different objectives. A constant encoding compresses maximally but has zero future information (counterexample D-E3-01); a predictively sufficient encoding may improve control and transfer in changing-mechanism environments (E3-H1, N0 conjecture, Section 8.6).

4.6 “High Subspace Alignment” Is Not Sufficient for Interface Qualification

The non-reversible circulation stress test shows that the dynamics–response direction cosine can be as high as 0.999 while response distortion, closure defect, and memory residual simultaneously cross pre-registered thresholds (Section 7.4 and Appendix D.3). High directional alignment can therefore coexist with joint interface admission failure.

4.7 Response Passports for Selective Causal Compression

The typed response kernel is used:

$$K_\tau^a(x, B) = P(F_{t+\tau} \in B \mid X_t = x, do(A_t = a)).$$

The response passport registers simultaneously:

$$\mathfrak{R} = (\mathcal{X}, \mathcal{A}, \mathcal{T}, \mathcal{F}, \{K_\tau^a\}, q, \mu, C, \mathcal{E}, Q).$$

Omitting task, environment, estimation, and quality fields before claiming a full response passport is forbidden. SCC-H1/H2/H3 are labeled conjectures (S grade) throughout this paper and must not be renamed theorems.

This section establishes four principles of conditional persistence and compression: persistence is contractual continued existence; formation and retention are separate accounts; four commutation relations differ; and an effective interface must pass five checks simultaneously. CP-T0, CP-T1, and C4I-T1 are conditional DCS results, the SCC series are conjectures, and open-system composition preservation is future work.

5 The Finite Exact CPIR Interface Quotient

This section is the central argument of the paper: under finite, exact, classical, frozen task contracts, event, history, state, and action quotients can coexist while preserving their respective types. Detailed proofs appear in Appendix B.10 (proof package C.21).

Why exact coexistence is not trivial. Physics has long known that quotienting a structure can be subtle even when the mathematics is exact. The representation theory of the inhomogeneous Lorentz group shows that symmetry quotients carry their own classification, not inherited automatically from the objects being quotiented (Wigner 1939); topological field configurations such as kinks obey statistics that survive only if the quotient respects them (Finkelstein and Rubinstein 1968); and in spacetime, the causal structure alone is enough to reconstruct the topology, which means any quotient that preserves causality must preserve topological facts as well (Malament 1977; Hawking et al. 1976). The lesson DCS takes into this section is that “coexistence” is a theorem to be proved premise by premise, not an assumption: event quotients, history-response quotients, state partitions, and action implementations each have their own preservation conditions, and the task contract is the frame that lets them coexist without one silently erasing another.

5.1 The F0-C Controlled Finite Model

Definition (finite classical CPIR system, CPIR-D1). A finite classical CPIR system consists of a finite event poset, a finite state space, a finite action set, and a finite history space; dynamics are finite row-stochastic kernels.

Definition (finite CPIR task contract, CPIR-D2). The finite task contract is:

$$\Theta = (\Pi, F, T, \mathcal{E}, d, \delta),$$

where Π is the admissible strategy set, F the future target, T the finite horizon, $\mathcal{E}$ the environment, d the distance, and δ the componentwise error budget. All strict results in this paper are stated under finite instance task contracts Θ ; the general specification Θ_r is not conflated with it.

5.2 Admissible Event Quotients and AEQ

Definition (DCS admissible event quotient, CPIR-D3). Let $q^E : E \twoheadrightarrow \bar{E}$ be a surjective map on the event set. The event quotient q^E maps the event set to a quotient event set and must satisfy the AEQ (admissible event quotient) condition: the transitive closure of the fiber graph of q^E is acyclic, so that the quotient relation $(\bar{E}, \prec_q)$ is again a finite strict poset. The state-abstraction map $\tau : X \twoheadrightarrow Z$ of Definition CPIR-D5 is a separate quotient type; it is not used to define the event quotient, and the two types must not be conflated.

Lemma (admissible event quotients preserve finite strict posets, CIPR-L1, conditional DCS result). Under AEQ, the transitive closure $(\bar{E}, \prec_q)$ of the fiber graph is a finite strict poset. The lemma reuses finite poset quotient results (Williams 2024); the DCS adaptation is to embed the event quotient into the task contract; full proof in Appendix B.10.

5.3 History-Response Quotients

Definition (exact history-response interface, CIPR-D4). For each lawful history h , the registered dynamics continue from its final state to generate futures, yielding the structural response kernel $K_{t:T}^\pi(h, \cdot)$. Define:

$$h \sim_\Theta h' \iff K_{t:T}^\pi(h, \cdot) = K_{t:T}^\pi(h', \cdot) \quad \forall \pi \in \Pi, 0 \leq t \leq T.$$

This equivalence relation yields the history-response quotient $\Phi^H : \mathcal{H} \rightarrow \mathcal{H}/\sim_\Theta$. Using structural response kernels avoids conditional distribution versions at zero-probability histories.

5.4 State Partitions and Strong Lumpability

Definition (per-implementation strong blockability, CIPR-D5). Let P^a be a stochastic kernel on X . For each action a and macro state z' :

$$S^a(x, z') = \sum_{y \in B_{z'}} P^a(x, y),$$

and assume:

$$\tau(x) = \tau(x') \implies S^a(x, z') = S^a(x', z') \quad \forall a, z'.$$

This is per-action strong blockability, reusing Kemeny–Snell/Buchholz strong lumpability (Kemeny and Snell 1976; Buchholz 1994).

Lemma (per-action strong blockability induces finite macro kernels, CIPR-L2, conditional DCS result). Under CIPR-D5, each registered micro action induces a unique macro kernel $\bar{P}^a$.

5.5 Multi-Implementation Invariance

Definition (macro-action multi-implementation invariance, CPIR-D6). Let $\Gamma : \bar{\mathcal{A}} \rightrightarrows \mathcal{A}$ satisfy $\Gamma(\bar{a}) \neq \emptyset$. Multi-implementation invariance requires:

$$S^a(x, z') = S^{a'}(x', z')$$

for any $\bar{a}$, $a, a' \in \Gamma(\bar{a})$, $\tau(x) = \tau(x')$, and z' . This guarantees that all micro implementations of the same macro action have identical macro effects.

Lemma (macro action kernels are well defined under multiple micro implementations, CPIR-L3, conditional DCS result). Under CPIR-D5/D6, the macro action kernel:

$$\bar{P}^{\bar{a}}(z, z') := S^a(x, z'), \quad x \in B_z, a \in \Gamma(\bar{a}),$$

is independent of the choice of state representative and implementation representative, and is a stochastic kernel on Z for each $\bar{a}$. Full proof in Appendix B.10.

5.6 Macro–Micro Finite-Horizon Response Commutation

The commutation statement below presupposes the implementation premises of Definitions CPIR-D5 and CPIR-D6, restated here because the result depends on them directly and not only on the appendix proof environment: (D5) each admissible micro action satisfies per-implementation strong blockability on the registered partition; (D6) the implementation correspondence $\Gamma : \bar{\mathcal{A}} \rightrightarrows \mathcal{A}$ is nonempty-covering and satisfies multi-implementation invariance. Without these two premises the macro kernel is not well defined and the commutation claim does not apply.

Lemma (finite-horizon macro–micro response commutation, CPIR-L4, conditional DCS result). Let the initial distribution be μ , and let the macro strategy be admissible, randomized, and dependent on the macro history via strategy kernels:

$$\bar{\pi}_t(\bar{a}_t \mid z_{0:t}, \bar{a}_{0:t-1}),$$

with micro implementations generated by compatible implementation kernels $\lambda_t(a_t \mid \bar{a}_t, h_t)$ whose support is contained in $\Gamma(\bar{a}_t)$. Then for every finite T :

$$\tau_{\#}^{0:T} P_{\mathfrak{C}}^{\mu, \pi} = P_{\bar{\mathfrak{C}}}^{\tau_{\#} \mu, \bar{\pi}}.$$

Here the push-forward $\tau_{\#}^{0:T}$ is defined on the state path space $X^{0:T}$; actions are transported jointly with states through the strategy–implementation protocol: the macro strategy $\bar{\pi}$ emits macro actions, and the compatible implementation kernels λ_t emit micro actions $a_t \in \Gamma(\bar{a}_t)$. The equality therefore holds for the joint state–action path distribution generated by the registered protocol, not merely for unconditional state marginals. If a micro controller changes the macro action distribution directly from compressed information, without decomposing into the strategy–implementation protocol above, the induction step fails.

If the target satisfies $F = \bar{F} \circ \tau^{0:T}$, then:

$$\mathcal{L}(F \mid \mu, \pi) = \mathcal{L}(\bar{F} \mid \tau_{\#}\mu, \bar{\pi}).$$

Conditions and boundaries. Path commutation does not require D7; D7 is used only to pass from path-distribution equality to equal response distributions for a specified target. Full proof in Appendix B.10.

5.7 The CPIR-T1 Theorem

Theorem (finite exact CPIR interface quotient, CPIR-T1, conditional DCS result). Suppose all six premises hold:

1. q^E satisfies AEQ;
2. Φ^H is an exact history-response quotient under Θ ;
3. τ satisfies D5 (per-implementation strong blockability) for every admissible micro implementation;
4. Γ nonempty-covering all macro actions and satisfies D6 (multi-implementation invariance);
5. targets, strategies, and initial distributions satisfy the L4 compatibility conditions;
6. if a structure-dynamics link is claimed, an implementation relation χ and its cross-quotient compatibility proof are additionally provided.

Then the interface package:

$$I_{\Theta} = (\bar{E}, \prec_q; Z, \bar{\mathcal{A}}, \{\bar{P}^{\bar{a}}\}; \mathcal{H}/\sim_{\Theta})$$

has five properties: (a) the quotient event relation is a finite strict poset; (b) macro action kernels are well defined; (c) finite-horizon path push-forward commutes for all registered compatible strategies and initial distributions, with D7 pushing the equality to target response distributions; (d) the history-response quotient preserves the responses registered in the task Θ ; (e) all four quotient types carry explicit lower-layer implementation fibers and effective domains. Full proof in Appendix B.10 (proof package C.21.6).

What cannot be concluded. The theorem is a modular stitching theorem on finite objects: it proves that four quotient types and maps can coexist within a common task contract while preserving their respective types. It does **not** derive Markov dynamics from event posets alone, does **not** prove that arbitrary partitions are automatically lawful (CPIR-X1), does **not** prove that the original certificate remains valid after SCC repair (CPIR-X2), and does **not** prove that compression automatically generates capabilities (CPIR-X8).

5.8 The C1P-C1 Corollary

Corollary (persistence-event probability preservation under the quotient, C1P-C1, conditional DCS result). Under the C1P-T1 conditions, suppose the persistence indicator satisfies:

$$G_T = \tilde{G}_T \circ \tau^{0:T},$$

then:

$$P_{\mathcal{E}}^{\mu, \pi}(G_T = 1) = P_{\mathcal{E}}^{\tau_{\#}\mu, \tilde{\pi}}(\tilde{G}_T = 1).$$

What cannot be concluded. Here “descends to the quotient” means that G_T factors through $\tau^{0:T}$; it does not mean that the persistence probability numerically decreases. If two in-block micro histories have different identity judgments, the equation $G_T = \tilde{G}_T \circ \tau^{0:T}$ fails and C1 cannot be invoked. The corollary does not apply to systems whose identity depends on compressed micro variables (C1P-X7).

5.9 Premise Necessity and Failure Exits

Removed condition	Failing stage	Corresponding counterexample
AEQ acyclicity	macro event relation may cycle	C1P-X1/X2
D5 strong blockability	macro kernel depends on in-block state	C1P-X3
D6 implementation invariance	one macro action has multiple macro effects	C1P-X4
D7 target measurability	paths commute but target still depends on compressed variables	C1P-X5
strategy–implementation compatibility	hidden micro information changes macro action distribution	C1P-X5 extension
finite horizon/closure	latent variables may bias later memory	C1P-X6
D8 persistence measurability	C1 identity events cannot be expressed at the macro level	C1P-X7
capability baseline and ablation	descriptive compression misread as capability growth	C1P-X8

6 The Finite Approximate CPIR Error Bound

This section gives CPIR-T2: a conditional error theorem for finite-state, discrete-time, classical Markov models, registered protocols, and total variation distance. Detailed proofs appear in Appendix B.11 (proof package C.22).

Why an error budget is the honest version of coarse graining. Whenever a macro description is not exactly autonomous, the question is not whether error exists but how it accumulates. The renormalization group made exactly this move in physics: scaling a system produces flows of couplings, and only the relevant directions survive (Wilson 1983); more recent work has made explicit that the RG can be read as an information-theoretic coarse graining in which relevance is measured by information loss (Gordon et al. 2021). In the same spirit, the geometric analysis of information loss under coarse graining shows that the discarded structure has a quantitative signature (Raju et al. 2018), and in field theory, approximate descriptions such as solitons are useful precisely because their errors are bounded and localizable (Skyrme 1961). Even discrete approaches to gravity treat spacetime itself as an approximation whose finite structure must be controlled rather than assumed away (Sorkin 2005). CPIR-T2 packages this shared intuition into a single finite-horizon certificate: a per-step joint defect, a contraction coefficient, and a telescoping bound that tells the user exactly how the macro–micro error grows, decays, or saturates.

6.1 Total Variation Distance

For probability distributions μ, ν on a finite set Z :

$$d_{\text{TV}}(\mu, \nu) = \frac{1}{2} \sum_{z \in Z} |\mu(z) - \nu(z)|.$$

The variational form gives, for $f : Z \rightarrow [0, 1]$, the bound $|\mathbb{E}_\mu f - \mathbb{E}_\nu f| \leq d_{\text{TV}}(\mu, \nu)$.

6.2 One-Step Joint Defect

Define the per-step defect between the micro-to-macro push-forward and the macro kernel:

$$\delta_t = \sup_{x \in X} d_{\text{TV}}(\tau_{\#} P_t(x, \cdot), \bar{P}_t(\tau x, \cdot)).$$

δ_t contains both the state-abstraction error and the macro-kernel error; it is a “joint defect”: whenever the macro–micro map does not commute exactly, δ_t is positive.

6.3 The Dobrushin Coefficient of the Macro Kernel

For a Markov kernel $\bar{P}_t$ on a finite set Z , define the Dobrushin contraction coefficient:

$$\alpha_t = \sup_{z, z' \in Z} d_{\text{TV}}(\bar{P}_t(z, \cdot), \bar{P}_t(z', \cdot)).$$

Lemma (Dobrushin contraction, external standard result). For any probability distributions μ, ν :

$$d_{\text{TV}}(\mu \bar{P}_t, \nu \bar{P}_t) \leq \alpha_t d_{\text{TV}}(\mu, \nu).$$

This lemma is a standard Markov contraction tool; DCS does not claim originality for it (Andrieux 2012).

6.4 One-Step Recursion

Let $e_t = d_{\text{TV}}(\tau_{\#} \mu_t, \bar{\mu}_t)$ be the macro–micro distribution error at step t . By the triangle inequality, the distribution-defect lifting lemma, and the contraction lemma:

$$e_{t+1} \leq \delta_t + \alpha_t e_t.$$

Intuition: the new error is at most the “one-step coupling defect” plus the contracted remainder of the old error under the macro kernel.

6.5 The Finite-Horizon Telescoping Bound

Expanding the one-step recursion yields:

$$e_T \leq e_0 \prod_{j=0}^{T-1} \alpha_j + \sum_{k=0}^{T-1} \delta_k \prod_{j=k+1}^{T-1} \alpha_j.$$

This decomposes the total-horizon error into initial-error decay plus a weighted accumulation of per-step defects, with weights determined by the later contraction coefficients.

6.6 The Contractive Bound and the Linear Bound

Theorem (finite approximate CIPR error theorem, CIPR-T2, conditional DCS result).

If $\alpha_t \leq \alpha < 1$ and $\delta_t \leq \delta$, then:

$$e_T \leq \alpha^T e_0 + \delta \frac{1 - \alpha^T}{1 - \alpha}.$$

If only $\alpha = 1$ is available (the macro kernel does not contract), only the linear finite-horizon bound remains:

$$e_T \leq e_0 + \sum_{t=0}^{T-1} \delta_t \leq e_0 + T\delta.$$

Conditions and boundaries. The theorem applies to finite-state, discrete-time, classical Markov models, registered total variation protocols, bounded per-step defects, and finite horizons. It does not apply to continuous, quantum, infinite-state, or strongly non-Markov cases; when $\alpha = 1$ there is no geometric decay and only the linear bound is available.

6.7 Macro-Measurable and Approximately Measurable Targets

If the macro target $f : Z \rightarrow [0, 1]$, then by the variational form:

$$|\mathbb{E}_{\tau_{\#}\mu_T} f - \mathbb{E}_{\bar{\mu}_T} f| \leq e_T.$$

If the micro target F is only approximately measurable: $\sup_x |F(x) - f(\tau x)| \leq \varepsilon_F$, then:

$$|\mathbb{E}_{\mu_T} F - \mathbb{E}_{\bar{\mu}_T} f| \leq \varepsilon_F + e_T.$$

What cannot be concluded. ε_F (target measurability error) must be registered separately from δ_t (dynamical defect) and must not be double-counted. If micro states in the same macro block have different targets or identities, ε_F can reach 1, and dynamical closure cannot substitute for a target measurability certificate (counterexamples CIPR-X5/X7 and the hidden target in Appendix C.9).

Scope of the persistence-event reading. The bound above is stated for terminal-state targets $F : X \rightarrow [0, 1]$, as registered in the task contract. Persistence indicators in this paper (Section 4) are generally history-dependent: the conditional persistence quantity G_T may depend on the whole history $X_{0:T}$, not only on the terminal state X_T . The T2 bound therefore inherits to a persistence event only in one of the following cases: (i) the persistence event is a terminal-state function of X_T ; or (ii) the required history has been encoded into an augmented state $Z_t = (X_t, M_t)$ under D2 memory augmentation (Section 2.5), and the augmented process satisfies the same finite Markov and contraction premises. Extending the bound to path-dependent persistence probabilities in general requires an augmented-state or path-space measure argument and is registered as future work (Section 9.4); CIPR-T2 does not claim automatic inheritance for all persistence events.

6.8 Source Identity and Machine Regression

Lemma 6.3 and the telescoping error recursion are standard Markov contraction/perturbation tools; Andrieux gives a published finite-time error bound for coarse-grained hidden Markov dynamics (Andrieux 2012). The DCS contribution of CPIR-T2 is to connect state abstraction, action implementation, response targets, and persistence events to the same type-safe contract. `models/cpir_approximate/` verifies the error bound time-step by time-step for the finite instance $\alpha = 0.5$, $\delta = 0.02$, and freezes the linear exit at $\alpha = 1$ and the non-measurable-target counterexample; machine regression only verifies implementation and boundary samples, and the F0 identity comes from finite mathematical proof (Appendix D.2).

This section established CPIR-T2: an explicit accumulated bound on the macro–micro distribution error over a finite horizon. It decays geometrically under contraction, leaving only the linear bound without contraction; macro-measurable targets inherit the state bound directly, and approximately measurable targets add a separate target error. The applicability domain is explicitly confined to finite, discrete-time, classical Markov models with total variation protocols.

7 Validation Analysis and Failure Exits

This section reports executable certificates within four classes of finite models. All numerical results are labeled “results within finite models”; they do not constitute empirical confirmation in natural systems. Complete parameters, seeds, and run commands appear in Appendix D.

What validation means for a certification framework. A framework whose main export is certification is only as credible as its failure exits: the same machinery that admits an interface must be able to reject one, and the rejection must be reproducible. This is the methodological counterpart of empirical disciplines that are built on registered failure — neuroscience, for example, quantifies collective dynamics through avalanche statistics precisely because deviations from the expected distribution are informative (Beggs and Plenz 2003), and the hierarchy of intrinsic timescales in cortex is inferred from data in which the null model is explicitly stated (Murray et al. 2014). In the same spirit, causal geometry treats the existence of a consistent causal structure as a falsifiable hypothesis about the data (Chvykov and Hoel 2021). The finite models in this section play that role for DCS 5.5: they are executable certificates — positive examples, mandatory counterexamples, boundary samples, and stress tests — not empirical evidence about nature, and the paper says so at every step.

7.1 Exact Positive Examples

`models/cpir_exact/` provides finite exact positive examples: on the registered partition, the strong blockability defect, target measurability defect, persistence measurability defect, and implementation invariance defect are all zero, and the macro kernel is well defined:

```
implementation_invariant: true
lumpability_defect: 0.0
macro_kernel: [[0.8, 0.2], [0.3, 0.7]]
persistence_measurability_defect: 0.0
target_measurability_defect: 0.0
```

Together with the CPIR-X1–X8 counterexamples, this positive example constitutes the machine regression gate for CPIR-T1/C1: passing these tests only unlocks the machine regression gate; the theorem identity is still jointly confirmed by the main-text proof, source mapping, and human review.

7.2 The CPIR-X1–X8 Counterexamples

The eight mandatory counterexamples kill the following wrong inferences:

ID	Inference killed	Key failing field
X1	arbitrary partitions automatically yield lawful poset quotients	macro_cycle: true
X2	certificates remain valid after SCC repair	requires_recertification: true
X3	arbitrary partitions automatically yield Markov macro kernels	lumpability_defect: 0.8
X4	macro actions naturally have unique micro-implementation effects	implementation_invariant: false
X5	dynamical closure automatically guarantees task sufficiency	target_measurability_defect: 1.0
X6	one-step fits automatically guarantee multi-step memorylessness	two_step_equal: false
X7	dynamical quotients automatically preserve persistence identity	persistence_measurability_defect: 1.0
X8	descriptive compression automatically generates new capabilities	new_reachable_capability: false

The complete “killed inference—minimal construction—failing field—repair exit” table for each counterexample appears in Appendix C.9.

7.3 T2 Numerical Execution Certificates

`models/cpir_approximate/` checks time step by time step for the finite instance $\alpha = 0.5$, $\delta = 0.02$ that the actual TV error does not exceed the geometric bound:

Horizon T	Actual error	Theoretical bound
1	0.02	0.02
4	0.0025	0.0375
8	0.000156	0.039844
12	0.000010	0.039990

Two boundary samples are also frozen: with $\alpha = 1$ only the linear finite-horizon bound is available (0.16 at $T = 8$); and with zero macro distribution error, a hidden, non-macro-measurable target can still reach maximal error (lesson: distribution closure does not certify a non-macro-measurable target). These samples verify implementation and counterexample responsibility, not empirical confirmation.

7.4 The Non-Reversible Stress Test

The non-reversible circulation family (SCC-F2) of `models/selective_compression_validation/` shows, under a scan of heterogeneity h , high directional alignment coexisting with joint admission failure. Pre-registered thresholds: response distortion at most 0.01, closure defect at most 0.005, two-step memory residual at most 0.005, dynamics–response direction cosine at least 0.99.

h	Response distortion	Closure defect	Memory residual	Direction cosine
0.0	0.0	0.0	0.0	1.0
0.1	0.00860	0.00400	0.00323	0.99998
0.2	0.01730	0.00800	0.00649	0.99993
0.8	0.07415	0.03200	0.02633	0.99889

The joint admission plateau endpoint is $h = 0.1$; at $h = 0.2$ all three errors cross the thresholds while the direction cosine remains as high as 0.99993. **Allowed conclusion:** high directional alignment can coexist with joint interface admission failure. **Forbidden conclusion:** a causal plateau in a real natural system has been empirically discovered.

7.5 The Cross-Scale Interface Joint Admission Prototype

`models/cross_scale_interface/` (CSR-F0-001) exhaustively enumerates all 15 partitions of the surviving states of a five-state, three-action model. In the symmetric model the registered partition $\{A0, A1\}|\{B0, B1\}|\{D\}$ simultaneously has zero within-class response distortion, zero closure defect, zero two-step memory residual, zero intervention commutation error, and zero persistence distortion (CSR-P1); at within-block heterogeneity 0.25, response, closure, memory, and persistence distortions turn positive while intervention commutation remains zero (CSR-P2), showing that the error axes are independent. The raw Pareto frontier is fully preserved, to exhibit that “one-block collapse” may remain on the frontier because it minimizes interface cost: Pareto membership itself does not constitute effective-layer certification.

7.6 Regression Gates

Appendix C.13 lists the mandatory regressions before freezing: D-CP-01, D-CP-02, D-ACTION-01, D-C4-01, D-RB-02, and D-CST-01 must pass; any unconditional occurrence in the main text of “EI implies stability,” “stationarity implies attraction,” “RAF implies life,” or “poset implies manifold” must fail the audit. The full project suite was rerun in the frozen Python 3.13 environment with 170/170 tests passing at RC6. Scope-directed CPIR tests continue to distinguish exact, approximate, and counterexample obligations; the expanded suite also guards registry, network, capability-passport, identity-transition, and release-manifest consistency without upgrading those scope-excluded materials into core preprint claims.

This section reports four classes of certificates within finite models: exact positive examples, the CPIR-X1–X8 counterexamples, T2 numerical execution certificates, and the non-reversible stress

test. Together they show that the framework can be defeated by counterexamples, error bounds can be verified numerically, and high alignment is not joint admission; all conclusions are confined to finite models.

8 Related Work, Originality, and Limitations

8.1 Comparison Method

Each comparison has four columns: inherited mature tools; DCS reorganization; candidate novelty; not yet proven. Conceptual juxtaposition is not originality, renaming is not derivation, and cross-domain coverage is not evidence. Self-archived preprints, reviews, theoretical models, numerical results, and direct experiments must be labeled separately.

8.2 Four-Column Comparison Table

Mature result	Applicability conditions	DCS interface	Remaining gap
Finite posets, Hasse diagrams, linear extensions (Bombelli et al. 1987; Norris 1997)	finite strict posets	AEQ contract for event quotients	infinite poset topology and measures; manifold embedding
Finite conservative Q, irreducible CTMC, KL contraction (Norris 1997; Kemeny and Snell 1976)	finite states, conservative, registered time	typed integration of stochastic semigroups into D0/D1/D2	non-normal generator spectra; growing-system limits
Strong lumpability (Kemeny and Snell 1976; Buchholz 1994)	fixed partition, strong blockability	CPIR-D5 per-implementation strong blockability	global spectral gap does not guarantee partition blockability
Causal abstraction (Rubenstein et al. 2017; Beckers and Halpern 2019; Beckers et al. 2020; Xia and Bareinboim 2025)	macro-micro intervention commutation	typed stitching of four maps under task contracts	latent variables and selection bias make targets non-identifiable
Computational mechanics/predictive equivalence (Crutchfield and Young 1989; Shalizi and Crutchfield 2001)	predictively sufficient states	E3 coding layer connected to individual boundaries	transferability under mechanism change

Mature result	Applicability conditions	DCS interface	Remaining gap
Information bottleneck (Tishby et al. 1999)	mutual information compression–relevance tradeoff	predictive vs causal compression separated	intervention semantics not preserved by mutual information
Mori–Zwanzig (Zwanzig 1961; Zwanzig et al. 1972; Mori 1965)	projection–memory structure	memory residual ϵ_{mem} registered	strongly coupled coarse-graining; non-Markov closure
Causal emergence/EI (Hoel et al. 2013)	specific intervention distributions	EI orthogonal to persistence (CP-T1)	general macro–micro EI gain theorem

8.3 Originality Boundaries

The originality of DCS should be stated only as: placing the mature results above into one typed task contract, requiring events, histories, states, interventions, persistence, and capabilities to be certified item by item, and providing a jointly failing interface package. **It must not be claimed** that DCS originated posets, lumpability, Markov contraction, causal abstraction, or the information bottleneck. When citing Zwanzig, the correction must be retained: only the projection–memory structure is inherited, not the uncorrected nonlinear transport conclusion (Zwanzig 1961; Zwanzig et al. 1972).

8.4 Currently Most Credible Original Combinations

1. a whole-architecture core–periphery failure isolation;
2. scientific governance via model-type registration, two-dimensional proposition grading, and retirement archives (Appendix E);
3. independent C4b closure and C5 persistence validation with final joint certification, and the type separation of geometric vs organizational coarse-graining;
4. E1 nested continuous individuality and E3 independent coding layers;
5. AI as a controlled laboratory rather than a physics proof proxy (RAI-H1, E0 conjecture, Section 8.6);
6. cross-domain “analogy license—bridging obligation—counterexample writeback” governance (Appendix E.5).

These are mostly framework-level innovations; they upgrade to scientific results only after producing new theorems, software, discriminating predictions, or independent empirical results.

Relationship to the wider cognitive, life, and AI research programs. The free-energy principle proposes that any adaptive system can be described as minimizing variational free energy, unifying perception, action, and learning under one normative account (Friston 2010); active inference gives that principle a process-level implementation in which agents select actions that realize preferred futures. DCS 5.5 does not compete with this program — it is strictly more modest: its task contracts provide a registration format in which claims about cross-scale causal interfaces (including, in principle, those made by free-energy and active-inference models) can be checked axis

by axis. Similarly, autopoiesis theory and the theory of major transitions define persistence and individuation at the level of living organization (Maturana and Varela 1980; Maynard Smith and Szathmary 1995), while the chemical origins literature treats the transition from prebiotic chemistry to gene regulation as a sequence of organizational thresholds (Flores-Garcia et al. 2026). Reinforcement learning, in turn, provides the operational definition of a task: an agent, a set of actions, a reward signal, and a horizon (Sutton and Barto 2018). Each of these traditions supplies one facet of the cross-scale question; none of them, on its own, provides the typed, failure-aware certification structure this paper proposes. Structural realism supplies the philosophical warning that motivates the whole enterprise: structure and implementation must be kept distinct if a theory is to survive theory change (Ladyman 1998).

8.5 Limitations

The strict results concentrate on finite-state models. Infinite poset topology and measures, identifiability of the native $\nu/\tau/J/U$ weights, non-explosion and local limits of growing systems, spectral analysis of non-normal generators, strongly coupled coarse-graining, non-Markov closure, and RG fixed points remain open. State spaces grow super-exponentially with node count; EI, full-partition search, and counterfactual interventions do not scale; metastable lifetimes can far exceed simulation windows. DCS currently has no general scale-existence theorem, and it has not proved that cross-domain scale indicators admit a common normalization. Two boundary literatures deserve explicit mention. First, the physics bridging branch (Section 9.4, item 5) concerns reconstructions of spacetime and symmetry from causal and geometric data — causal set theory (Sorkin 2005), order- and number-based reconstruction programs (Braun 2026; Braun and Sämann 2025), and Lorentzian metric-space convergence (Minguzzi and Suhr 2024), as well as generalized global symmetries in field theory (Gaiotto et al. 2015); none of these is implied by the finite results of this paper, and the paper does not claim otherwise. Second, non-Markovianity is increasingly a measurable, controlled property rather than a nuisance: process-tensor methods characterize and control non-Markovian quantum processes experimentally (White et al. 2020), which is exactly why the D2 memory-augmentation registration of Section 2.5 is a design choice and not a workaround.

8.6 Conditional Conjectures

The following are N0/N1/E0/S-grade conjectures and must not be written as universal theorems in the abstract:

- some dynamical systems have cross-sample-stable scale–organization Pareto frontiers (C4-H1);
- predictively sufficient encodings can improve control and transfer in changing-mechanism environments (E3-H1);
- controllable AI systems can test parts of the C4/C5/E1/E3/E4 universal propositions (RAI-H1, E0);
- social variables may have independent predictive gain for some brain measures after controlling for diet, ecology, body mass, and phylogeny (RNE-H1, restricted);
- SCC-H1/H2/H3 are candidate conjectures for conditional causal plateaus, dynamics–response RG alignment, and interface regeneration; this paper does not rename them theorems.

8.7 Theory Termination Conditions

If the core type system cannot produce fewer errors or stronger predictions than existing tools; if scale–organization methods show no stable gain across models; if all R branches have no discriminating predictions over a long horizon, then DCS should be reduced to a methodological framework rather than expanded into a unified theory. Termination, contraction, or downgrade is also a scientific result.

The original contribution of DCS is typed stitching and failure-exit governance, not the invention of foundational theorems; mature results, DCS interfaces, and remaining gaps are compared item by item in the four-column table. Limitations concentrate on the finite scope, conjectures retain their conjecture identity, and termination conditions are part of the theory itself.

9 Conclusion and Outlook

9.1 What This Paper Strictly Establishes

Under finite-state, discrete-time, classical Markov models and frozen task contracts, this paper establishes:

1. **CPIR-T1**: under six explicit premises, event, history, state, and action quotients coexist in one typed interface package while preserving their respective types;
2. **CPIR-C1**: the probability of quotient-measurable persistence events is preserved under the exact quotient;
3. **CPIR-T2**: an explicit finite-horizon accumulated error bound in total variation distance, decaying geometrically under contraction and leaving only the linear bound otherwise;
4. **CP-T0 and CP-T1**: the survivor conditioning enrichment identity and the persistence–EI four-quadrant finite counterexample;
5. **C4I-T1**: composition preservation of two-level exact causal abstractions.

These conclusions carry the identity “conditional DCS result” or “inherited external standard result,” and each is registered in the proposition ledger of Appendix E.1.

9.2 What This Paper Only Provides: Finite-Model Certificates

The numerical part of this paper provides executable certificates within finite models: exact positive examples, the CPIR-X1–X8 mandatory counterexamples, the T2 error-bound execution certificates, selective-compression cross-model validation with the non-reversible stress test, and the cross-scale interface joint admission prototype. These certificates show that the framework can be defeated by counterexamples, error bounds can be verified numerically, and high directional alignment is not joint admission; they do not constitute empirical confirmation in natural systems.

9.3 What Remains Conjecture

The following retain conjecture identity and must not be renamed theorems:

- the general existence of conditional causal plateaus (SCC-H1/H2/H3);
- cross-sample stability of scale–organization Pareto frontiers (C4-H1);
- predictively sufficient encodings improving control and transfer (E3-H1);
- AI as a strong intervention laboratory (RAI-H1);
- a positive association between causal emergence and persistence (CE-H1).

9.4 What Is Future Work

The following are explicitly registered as future work in this paper; they are not written into the “completed contributions” of the abstract, nor stated as theorems of this paper:

1. **P2B finite interconnection prototype:** general composition preservation for open interconnected systems requires finite interconnection positive examples and mandatory counterexamples; this preprint does not claim completion;
2. **assume–guarantee contracts for open systems:** extending closed task contracts to open systems with environment interfaces;
3. **the life dual contract and persistence-carrier transitions:** life admission distinguishes sustained reactions, RAF, dynamical self-maintenance, cross-temporal constraint closure, individuation, and evolutionary individuals, and is not completed;
4. **the intelligence capability lattice and collective agency:** intelligence measures are task-choice dependent, and the origin of goals and normative values lies beyond dynamical models;
5. **5.6 physics bridging:** the RQ/RST/RP branches are the strongest-vision, weakest-evidence region and must not become the center of current academic promotion; deriving four-dimensional Lorentzian spacetime, quantum field theory, or the Standard Model from DCS is not completed;
6. **6.0 discriminating empirical predictions:** no 6.0-grade discriminating empirical prediction currently exists;
7. **path-space and augmented-state error bounds:** extending the CPIR-T2 bound from terminal-state targets to history-dependent persistence events requires an augmented-state or path-space measure argument (Section 6.7), which is not claimed by this paper.

9.5 The Final Boundary

This paper is a formal research draft, not a finished theory of everything. DCS 5.5 provides a finite, type-safe, auditable, counterexample-defeatable certification framework for cross-scale causal interfaces, and states clearly what it has not proved. Its near-term success criteria are clear theorems, reproducible tools, and fallible experiments, not interpretive vocabulary that covers every scale.

Outlook. The framework is deliberately positioned at the junction of three ongoing research programs rather than inside any one of them: cognitive science, where free-energy and active-inference accounts make strong cross-scale claims about brains that DCS can register as task contracts rather than assert (Friston 2010); evolutionary theory, where major transitions show that persistence carriers migrate and re-certify at each new level (Maynard Smith and Szathmary 1995); and AI, where reinforcement learning operationalizes exactly the kind of frozen task contract this paper uses (Sutton and Barto 2018). If the finite results here hold up, the next steps are the open-interconnection prototype (P2B) and discriminating empirical predictions; if they do not, the failure exits are already registered. Either outcome is progress, which is the sense in which DCS 5.5 is a certification framework first and a theory second.

The conclusion closes in four layers: this paper strictly establishes finite exact quotients and approximate error bounds; it provides only finite-model certificates; SCC and cross-scale conjectures

remain conjectures; and P2B, life, intelligence, physics bridging, and 6.0 predictions are all registered as future work. The final positioning is “typed certification framework plus explicit failure exits,” not “an explanation of everything.”

Appendix A Symbols, Types, and Model Registration

A.1 Type Table for Events, States, Histories, Actions, Dynamics, and Observables

One symbol, one type within the same chapter is mandatory. If R is used as a branch prefix (RQ/RST etc.), it must not be conflated with lifted kernels.

Symbol	Type	Meaning
V	set	event/node set
$\prec$	relation on $V \times V$	strict poset
ν	$\mathcal{A}_V \rightarrow [0, \infty]$	region volume/sampling measure
τ	$\{(x, y) : x \preceq y\} \rightarrow [0, \infty)$	candidate Lorentz time separation
J	registered relation/interval to reals, matrices, or operators	interaction coupling kernel
$U_{y \leftarrow x}$	group representation $F_x \rightarrow F_y$	forward parallel transport
Φ	section/field over events, relations, or intervals	optional matter field or internal field
ω	element of Ω	one lawful configuration
Ω, Σ	measurable spaces	configuration space and event family
f	element of $\mathcal{F}(\Omega)$	observable function
μ, π	elements of $\mathcal{P}(\Omega)$	probability measure, stationary measure
$q(\omega, \omega')$	$\Omega \times \Omega \rightarrow [0, \infty)$	continuous-time configuration jump rate
$S[C]$	history/full configuration to $\mathbb{R}$	action functional
$D(A, B)$	history event pairs to $\mathbb{C}$	decoherence functional
L	operator	backward generator
P_t	operator semigroup	transition semigroup
C	map	micro-to-macro coarse-graining
$C_\#$	measure map	push-forward
R	Markov kernel	macro-to-micro lifting
X, B, E	stochastic processes	internal, boundary, environment
H, Z	random variables	history and encoding state
A	set/variable	action

Symbol	Type	Meaning
Causal-semantics extension:		
Symbol	Type	Meaning
$\mathcal{I}$	intervention set	operations allowed by the model
ϑ	intervention map $\mathcal{I}_M \rightarrow \mathcal{I}_{\bar{M}}$	micro-to-macro intervention; typed separately from configuration variables
$A = C^{-1}(\bar{A})$	measurable set	candidate pattern identity class
τ_{A^c}	stopping time	first exit from the identity class
$S_A(T)$	$[0, 1]$	finite-time survival probability
$b_A(t)$	nonnegative function	formation intensity
ε_{dyn}	nonnegative real	dynamical closure residual
ε_{int}	nonnegative real	macro–micro intervention commutation error
C_{maint}	registered-dimension cost	maintenance energy, matter, or computation cost

A.2 Domains and Codomains of the Four Maps

Map	Admissible input	Admissible output	Primary admission
q_r^E	events and event relations	event quotient	poset contract and structure preservation
Φ_r^H	full or truncated histories	interface/memory state	task response sufficiency
τ_r^X	micro states/state distributions	macro states/pushed-forward distributions	dynamical closure and representation error
Γ_r^A	macro actions	sets of micro implementations	implementability and implementation invariance

Γ_r^A is a set-valued implementation correspondence, not a single-valued function. Push-forward, pull-back, and conditional expectation projection have different directions: measure push-forward is written $C_{\#}\mu$; function pull-back is $f \circ C$; conditional expectation projection is $Pf = \mathbb{E}[f \mid \sigma(C)]$. Geometric coarse-graining is written C_b^{geo} , organizational state partition C^{org} ; state maps, statistical estimators, and physical interpreters must not share the same C without annotation.

A.3 Model Families and Profiles

Model families are F0/F1/G/O/A/R: F0 finite exact; F1 proof sketch or not yet formalized; G growing systems; O open organizations; A action/agent; R representation branches (RQ/RST/RP etc.). CIPR profiles are CIPR-C (full certification), CIPR-Q (quantitative scoring), PRE-CIPR (pre-certification research). Switching from F0 to F1, G, O, A, or R requires a new model card and re-validation of state space, regularity, non-explosion, boundaries, and intervention premises; the original theorem numbering must not be reused.

A.4 General and Finite Instance Task Contracts

The general task contract is registered at scale r : $\Theta_r = (\mathcal{A}_r^\Theta, F_r, T_r, \mathcal{E}_r, d_r, \epsilon_r)$. The finite instance: $\Theta = (\Pi, F, T, \mathcal{E}, d, \delta)$. They are a general specification and a finite instance; their similar symbols must not lead to merging fields.

A.5 E/Q/K and the C5 Passport

Error vector $\mathbf{E}_r = (\epsilon_{ord}, \epsilon_{dyn}, \epsilon_{resp}, \epsilon_{impl}, \epsilon_{int}, \epsilon_{mem})$; compression gain vector $\mathbf{Q}_r = (R_{state}, R_{history}, R_{parameter}, R_{co})$; capability result vector $\mathbf{K}_r = (\mathcal{R}_{reliable}, C_{ctrl}, T_{pred}, R_{robust}, G_{env})$. The three tables use itemwise thresholds, confidence intervals, and Pareto relations; any aggregate score must attach weight sources and sensitivity analysis. The C5 persistence passport (**PersistenceClaim**) registers object, identity, model, scale, time window, environment, perturbation, indicator, cost, null model, evidence, and failure conditions, and must not be automatically derived from any single E/Q/K vector.

A.6 Preconditions Checklist for Theorem Use

1. both sides of an equality lie in the same space or have an explicit isomorphism;
2. domains and codomains of composed operators match;
3. nonlinear coarse-graining is not written as a linear projection;
4. push-forward, pull-back, and conditional expectation directions are correct;
5. generators satisfy the positive maximum principle/conservativity;
6. dimensions are consistent;
7. classical probability and quantum amplitudes are not mixed;
8. graph spectral types are explicit.

Appendix A gives the full symbol type table of the main text, the domains and codomains of the four maps, model family and profile classification, the two task contract forms, the three scorecards with the persistence passport, and an eight-item type audit checklist before theorem use.

Appendix B Full Derivations and Proofs

This appendix collects all proofs invoked by the main text. Each proof is labeled by source identity: inherited external standard result or conditional DCS result. The F0 label means a conventional mathematical proof is complete and has been reviewed by a human; it does not mean Lean/Coq is complete; proof-assistant status follows the `verification` field of the registry.

B.1 C1-T1: Every Finite Poset Has a Linear Extension (inherited external standard result)

Theorem C1-T1. Every nonempty finite poset has a minimal element and admits a linear extension.

Proof. Take any x_0 . If it is not minimal, take $x_1 \prec x_0$; if x_1 is still not minimal, continue with $x_2 \prec x_1$. Strictness guarantees that the elements of the chain are distinct, and finiteness forbids an infinite distinct descending chain; hence the process terminates at some minimal element m .

Delete m from the poset; the restricted relation is still a finite poset. Induct on element count n : the empty set is trivial; assuming all posets with fewer than n elements admit linear extensions, take a minimal element m of an n -element poset, linearly extend the remaining $n - 1$ elements, and place m first. Since no $x \prec m$ exists and the remaining relations are preserved by the inductive order, the resulting total order preserves the original poset.

Implementation obligation: “zero in-degree” in programs must be computed on the Hasse diagram or on a DAG consistent with the transitive closure; inputs with cycles must be rejected before sorting.

B.2 C1-T2: Transitive Reduction of a Finite DAG Is Unique (inherited external standard result)

Theorem C1-T2. The transitive reduction of a finite DAG is unique and equals the Hasse diagram of its reachability poset.

Proof. Let the finite DAG be $G = (V, E)$, and write $a \prec b$ for the existence of a nonempty directed path from a to b . Acyclicity makes $\prec$ a strict poset. Let H contain only the cover edges of this poset.

First prove that the transitive closure of H equals $\prec$. If $a \prec b$, consider the finite interval $[a, b] = \{x \in V : a \preceq x \preceq b\}$. If a does not cover b , take a strict intermediate element; repeated refinement must terminate because the interval is finite, yielding a cover chain from a to b . Hence every

reachable pair lies in the closure of H ; the reverse inclusion follows since cover edges are themselves in $\prec$.

Now prove minimality and uniqueness. Let $G' = (V, E')$ have the same transitive closure as G . If a covers b , any path from a to b in G' cannot contain two or more edges, since the first intermediate vertex x would satisfy $a \prec x \prec b$, contradicting coverage. Hence $a \rightarrow b$ must be a direct edge of E' . Any graph with the same closure must contain all Hasse edges, and H itself already has that closure; therefore H is the unique minimal edge set.

B.3 C2-P1: Lawful Rule Sequences Preserve Closure (conditional DCS result)

Let $r_k : D_k \subseteq \Omega \rightarrow \Omega$ all be lawful, i.e., $r_k(D_k \cap \Omega) \subseteq \Omega$. Induct on the length n of an executable sequence $r_n \circ \dots \circ r_1$. For $n = 0$ the identity map preserves Ω . If the first n steps yield $\omega_n \in \Omega$ and $\omega_n \in D_{n+1}$, lawfulness gives $r_{n+1}(\omega_n) \in \Omega$. Hence any finite executable lawful rule sequence preserves the closure of the configuration space.

The proof guarantees neither that an executable rule always exists, nor that rewriting terminates, nor that continuous-time processes are non-explosive.

B.4 C3-T1: Stochastic Semigroup of a Finite Conservative Q (inherited external standard result)

Let $Q = (q_{ij})$ be a finite conservative Q matrix with departure rates $q_i = -q_{ii}$. Take $\lambda > \max_i q_i$ and $K = I + Q/\lambda$. The strict inequality makes every diagonal entry of K positive; for positivity of the semigroup it already suffices to take any $\lambda \geq \max_i q_i > 0$. Off-diagonal entries are nonnegative, diagonal entries are nonnegative, and every row sums to one, so K is a stochastic matrix. Since $Q = \lambda(K - I)$ and K commutes with I ,

$$e^{tQ} = e^{-\lambda t} e^{\lambda t K} = e^{-\lambda t} \sum_{m=0}^{\infty} \frac{(\lambda t)^m}{m!} K^m.$$

The right-hand side is a convex mixture, with Poisson weights of parameter λt , of the stochastic matrices K^m ; hence $P_t = e^{tQ}$ has nonnegative entries and unit row sums. The matrix exponential identity gives $P_0 = I$ and $P_{s+t} = P_s P_t$; existence and uniqueness of the global solution of a finite-dimensional linear ODE give the forward and backward equations.

B.5 C3-T2: Stationarity and Convergence of a Finite Irreducible CTMC (inherited external standard result)

Theorem C3-T2. A finite CTMC has at least one stationary distribution; if its transition graph is irreducible, the stationary distribution is unique, strictly positive, and every initial distribution converges to it as $t \rightarrow \infty$.

Proof. Every finite directed transition graph contains at least one closed communicating class. On that class take $\lambda > \max_i q_i$ and uniformize to $K = I + Q/\lambda$. The restriction of K to the closed class is a finite irreducible stochastic matrix, hence has a unique strictly positive stationary distribution; extending it by zero outside the class gives a stationary distribution of the original chain. Thus any finite chain has at least one stationary distribution.

If the chain is irreducible, K is irreducible; the strict inequality also gives each state a positive self-loop probability, so K is aperiodic. The Perron–Frobenius theorem for finite irreducible aperiodic stochastic matrices gives a unique strictly positive probability vector $\pi K = \pi$ with $K^m \rightarrow \mathbf{1}\pi$. Since $Q = \lambda(K - I)$, we have $\pi Q = 0$, so π is stationary for the CTMC.

Use the induced row total-variation norm $\|\cdot\|_{\infty,TV}$. For any $\varepsilon > 0$ take M such that $\|K^m - \mathbf{1}\pi\|_{\infty,TV} < \varepsilon$ for all $m \geq M$. By the Poisson mixture decomposition of B.4,

$$\|P_t - \mathbf{1}\pi\|_{\infty,TV} \leq \Pr\{N_t < M\} + \varepsilon, \quad N_t \sim \text{Poisson}(\lambda t).$$

The first term tends to zero as $t \rightarrow \infty$, hence $P_t \rightarrow \mathbf{1}\pi$, and therefore $\mu P_t \rightarrow \pi$. Uniqueness follows by taking the limit of any stationary distribution.

B.6 C3-T3: Relative Entropy Contraction (inherited external standard result)

Any Markov kernel P satisfies the data-processing inequality $D(\mu P \| \nu P) \leq D(\mu \| \nu)$, with relative entropy understood in the extended-real sense; if the right-hand side is finite, then $\mu \ll \nu$. Take $\nu = \pi$ with $\pi P_t = \pi$, giving $D(\mu P_t \| \pi) \leq D(\mu \| \pi)$. For $0 \leq s < t$, apply the same inequality to the initial distribution μP_s and the kernel P_{t-s} , yielding that relative entropy is nonincreasing along the semigroup. This is the Markov data-processing property and does not automatically equal a law of thermodynamic entropy production.

B.7 C4-T1: Strong Lumpability (inherited external standard result)

Let the finite state space be partitioned by $C : \Omega \rightarrow \bar{\Omega}$, and let $U_C \bar{f} = \bar{f} \circ C$ pull macro functions back to block-constant functions. If any two states $i, j \in B_\alpha$ in the same source block have equal total rates into every target block B_β , define

$$\bar{q}_{\alpha\beta} = \sum_{k \in B_\beta} q_{ik}, \quad i \in B_\alpha.$$

The strong lumpability condition guarantees that the right-hand side is independent of the representative i ; off-diagonal terms are nonnegative and row sums are zero, so $\bar{Q}$ is a conservative macro generator. For any macro function $\bar{f}$ and $i \in B_\alpha$,

$$(LU_C \bar{f})(i) = \sum_{\beta} \bar{q}_{\alpha\beta} [\bar{f}(\beta) - \bar{f}(\alpha)] = (U_C \bar{L} \bar{f})(i).$$

This gives the generator commutation $LU_C = U_C\bar{L}$. By induction $L^n U_C = U_C\bar{L}^n$, and summing term by term in the matrix exponential for finite-dimensional matrices yields $P_t U_C = U_C \bar{P}_t$. Pairing with any micro probability measure μ and macro observable $\bar{f}$ gives the dual relation $C_\# P_t^* = \bar{P}_t^* C_\#$.

B.8 CP-T0: Survivor Conditioning Enrichment Identity (conditional DCS result)

Let Z be a pattern feature measured at formation, and $S_T = \{\tau_{Ac} > T\}$ the event of surviving to time T under a pre-registered identity contract. For any measurable set B , if $\Pr(Z \in B) > 0$ and $\Pr(S_T) > 0$, then

$$\Pr(Z \in B \mid S_T) = \frac{\Pr(S_T \mid Z \in B) \Pr(Z \in B)}{\Pr(S_T)}.$$

With dominating measure μ , if Z has initial density $p_0(z)$ with respect to μ , and $S_z(T) = \Pr(\tau_{Ac} > T \mid Z = z)$, then, whenever the relevant regular conditional distributions exist,

$$p_T(z \mid \tau_{Ac} > T) = \frac{p_0(z) S_z(T)}{\int p_0(z') S_{z'}(T) \mu(dz')}.$$

Proof. By the multiplication rule of conditional probability, $\Pr(Z \in B, S_T) = \Pr(S_T \mid Z \in B) \Pr(Z \in B) = \Pr(Z \in B \mid S_T) \Pr(S_T)$. Dividing both sides by the positive number $\Pr(S_T)$ gives the first identity; applying the same identity at the point mass z and normalizing by the law of total probability gives the second. In discrete feature spaces the integral is a sum.

Corollary. If $\Pr(S_T \mid Z \in B) > \Pr(S_T)$, then $\Pr(Z \in B \mid S_T) > \Pr(Z \in B)$. This corollary only states composition change under conditional sampling. It neither identifies a causal effect of Z on survival, nor implies that the system actively maximizes Z , lifetime, or any scalar “persistence potential.”

B.9 CP-T1: Finite Models Realize the Persistence–EI Four Quadrants (conditional DCS result)

Let $n \geq 2$, and take pairwise disjoint finite sets $A = \{a_1, \dots, a_n\}$, $B = \{b_1, \dots, b_n\}$. The system starts from the uniform intervention $I \sim \text{Unif}(A)$ on A ; let $Y = X_1$ be the one-step output. One-step effective information is $\text{EI}_1 = I(I; Y)$ (base-2 logarithm). The exit time and persistence probability of the candidate pattern are

$$\tau_{Ac} = \inf\{t \in \mathbb{N}_0 : X_t \notin A\}, \quad S_A(T) = \Pr(\tau_{Ac} > T).$$

Define four deterministic transition kernels:

$$\begin{aligned} F_{HH}(a_i) &= a_{i+1 \pmod n}, & F_{HL}(a_i) &= a_1, \\ F_{LH}(a_i) &= b_i, & F_{LL}(a_i) &= b_1. \end{aligned}$$

Here the cyclic index convention is $a_{n+1} = a_1$. Completing the transitions of states in B arbitrarily yields lawful finite discrete-time Markov kernels; the following conclusions depend only on the one-step transitions from A .

Theorem CP-T1. The four kernels realize, respectively, “high persistence–high EI,” “high persistence–low EI,” “low persistence–high EI,” and “low persistence–low EI”:

	$S_A(1)$	EI_1
F_{HH}	1	$\log_2 n$
F_{HL}	1	0
F_{LH}	0	$\log_2 n$
F_{LL}	0	0

Proof. For F_{HH} and F_{HL} , all images of the a_i remain in A , so $\tau_{Ac} > 1$ and $S_A(1) = 1$. Iterating these transitions on A keeps the state in A , so $S_A(T) = 1$ for all integers $T \geq 0$. For F_{LH} and F_{LL} , all a_i enter B at the first step, so $\tau_{Ac} = 1$ and $S_A(1) = 0$.

All four kernels are deterministic, so $H(Y | I) = 0$ and by definition $\text{EI}_1 = I(I; Y) = H(Y) - H(Y | I) = H(Y)$. F_{HH} is a bijection on A , and F_{LH} is a bijection from A to B ; uniform input therefore produces n equally likely outputs, giving $H(Y) = \log_2 n$. F_{HL} and F_{LL} are constant maps producing a single output, so $H(Y) = 0$. All four claims hold.

Therefore, within a finite model class with fixed state space, candidate identity set, time window, and intervention distribution, persistence neither implies high one-step EI nor is implied by it. The theorem only negates logical identification of the two indicators; it does not negate statistical association in specific empirical model families, nor does it equal a macro–micro EI gain theorem.

B.10 The CPIR-T1 Proof Package (conditional DCS result)

This section proves CPIR-L3, CPIR-L4, CPIR-T1, and CPIR-C1 of 09E. The proof reuses mature results of finite poset quotients and finite Markov strong blockability; the new DCS content is the typed stitching of multi-implementation actions, history-dependent strategies, response targets, event quotients, and persistence events.

B.10.1 Model, Notation, and Strategy Compatibility

Let $X, Z, \mathcal{A}, \bar{\mathcal{A}}$ be nonempty finite sets, $\tau : X \twoheadrightarrow Z$ surjective, and write $B_z = \tau^{-1}(z)$. For each $a \in \mathcal{A}$, let P^a be a stochastic kernel on X , and define $S^a(x, z') = \sum_{y \in B_{z'}} P^a(x, y)$. Let $\Gamma : \bar{\mathcal{A}} \rightrightarrows \mathcal{A}$ satisfy $\Gamma(\bar{a}) \neq \emptyset$. Assume:

$$\tau(x) = \tau(x') \implies S^a(x, z') = S^a(x', z') \quad \forall a, z',$$

and define the multi-implementation invariance condition as $S^a(x, z') = S^{a'}(x', z')$ for any $\bar{a}$, $a, a' \in \Gamma(\bar{a})$, $\tau(x) = \tau(x')$, and z' . The former is per-action strong blockability; the latter is multi-implementation invariance.

Define admissible randomized macro-history-dependent strategy kernels:

$$\bar{\pi}_t(\bar{a}_t \mid z_{0:t}, \bar{a}_{0:t-1}).$$

Compatible implementations generate actions by a two-stage protocol: first the strategy kernel selects $\bar{a}_t$, then the implementation kernel $\lambda_t(a_t \mid \bar{a}_t, h_t)$ (with $\text{supp } \lambda_t(\cdot \mid \bar{a}_t, h_t) \subseteq \Gamma(\bar{a}_t)$) selects the micro action. Since h_t may contain the full micro history, this contract does not require the implementer to observe only the macro state.

B.10.2 CPIR-L3: Macro Action Kernels Are Well Defined

Lemma CPIR-L3. Under the above conditions, $\bar{P}^{\bar{a}}(z, z') := S^a(x, z')$ (for $x \in B_z$, $a \in \Gamma(\bar{a})$) is independent of the choice of state representative x and implementation representative a , and is a stochastic kernel on Z for each $\bar{a}$.

Proof. Fix $z, z', \bar{a}$. Take any two admissible representatives $x, x' \in B_z$ and $a, a' \in \Gamma(\bar{a})$. Multi-implementation invariance directly gives $S^a(x, z') = S^{a'}(x', z')$, so the macro kernel is independent of both representative choices. Nonnegativity is inherited from P^a . Since $\{B_{z'} : z' \in Z\}$ is a partition of X , $\sum_{z' \in Z} \bar{P}^{\bar{a}}(z, z') = \sum_{z' \in Z} \sum_{y \in B_{z'}} P^a(x, y) = \sum_{y \in X} P^a(x, y) = 1$. Hence the macro kernel is a well-defined stochastic kernel.

B.10.3 One-Step Conditional Commutation

Fix a micro history h_t with final state x_t and macro final state $z_t = \tau(x_t)$. Given a macro action $\bar{a}_t$, the micro action is chosen by the compatible implementation kernel. Then

$$P(\tau(X_{t+1}) = z' \mid h_t, \bar{a}_t) = \sum_{a \in \Gamma(\bar{a}_t)} \lambda_t(a \mid \bar{a}_t, h_t) S^a(x_t, z') = \sum_{a \in \Gamma(\bar{a}_t)} \lambda_t(a \mid \bar{a}_t, h_t) \bar{P}^{\bar{a}_t}(z_t, z') = \bar{P}^{\bar{a}_t}(z_t, z').$$

Hence any micro-history-dependent implementation choice produces the same macro one-step kernel after push-forward. No assumption of uniform within-block micro conditional distributions is made.

B.10.4 CPIR-L4: Finite-Horizon Path and Response Commutation

Lemma CPIR-L4. Let the initial distribution be μ , the macro strategy be admissible, randomized, and macro-history-dependent, and micro implementations satisfy the compatibility condition. Then for every finite T :

$$\tau_{\#}^{0:T} P_{\mathfrak{C}}^{\mu, \pi} = P_{\bar{\mathfrak{C}}}^{\tau_{\#} \mu, \bar{\pi}}.$$

If the target satisfies $F = \bar{F} \circ \tau^{0:T}$, then $\mathcal{L}(F \mid \mu, \pi) = \mathcal{L}(\bar{F} \mid \tau_{\#}\mu, \bar{\pi})$.

Proof. Induct on t to show that the finite-dimensional distributions of macro state–action histories coincide. At $t = 0$, $P(\tau(X_0) = z) = \sum_{x \in B_z} \mu(x) = (\tau_{\#}\mu)(z)$, so the initial macro distributions agree. Assume agreement up to macro history at time t . Given any macro history $z_{0:t}, \bar{a}_{0:t-1}$, both systems select $\bar{a}_t$ by the same kernel $\bar{\pi}_t$. In the micro system, even if the within-block micro history distribution conditional on that macro history is arbitrary, the one-step conditional commutation still guarantees that the next macro state distribution is $\bar{P}^{\bar{a}_t}(z_t, \cdot)$, identical to the macro system’s transition rule. By the law of total probability, the length- $(t + 1)$ macro state–action history distributions agree. Induct to T and marginalize over action histories to obtain path push-forward commutation. Finally, if $F = \bar{F} \circ \tau^{0:T}$, then for any set A in the outcome space, $P(F \in A) = P(\tau^{0:T}(X_{0:T}) \in \bar{F}^{-1}(A)) = P_{\bar{\mathcal{E}}}^{\tau_{\#}\mu, \bar{\pi}}(\bar{F}(Z_{0:T}) \in A)$, so response commutation holds.

Path commutation does not require D7; D7 is used only to pass from path-distribution equality to equal response distributions for a specified target. If a micro controller changes the macro action distribution directly based on compressed information without decomposing into the strategy–implementation protocol, the induction step fails.

B.10.5 Well-Definedness of the History-Response Quotient

For each lawful history h , the registered dynamics continue from its final state to generate futures, yielding the structural response kernel $K_{t:T}^{\pi}(h, \cdot)$. Define $h \sim_{\Theta} h'$ iff $K_{t:T}^{\pi}(h, \cdot) = K_{t:T}^{\pi}(h', \cdot)$ for all strategies and times. Equality in the space of probability measures is reflexive, symmetric, and transitive; universal quantification over strategies and times preserves these three properties, so $\sim_{\Theta}$ is an equivalence relation and $\Phi^H : \mathcal{H} \rightarrow \mathcal{H}/\sim_{\Theta}$ is well defined. Using structural response kernels avoids conditional distribution versions at zero-probability histories.

B.10.6 CPIR-T1: The Finite Exact Interface Quotient

Theorem CPIR-T1. Under the six admission conditions, the interface package

$$I_{\Theta} = (\bar{E}, \prec_q; Z, \bar{\mathcal{A}}, \{\bar{P}^{\bar{a}}\}; \mathcal{H}/\sim_{\Theta})$$

has five properties.

Proof. Verify item by item:

1. By the AEQ condition and CPIR-L1, the transitive closure $(\bar{E}, \prec_q)$ of the fiber graph is a finite strict poset.
2. By per-action strong blockability, CPIR-L2 gives the macro kernel induced by each micro action; by CPIR-L3 these kernels are independent of state and implementation representatives within each $\Gamma(\bar{a})$, so $\{\bar{P}^{\bar{a}}\}$ is well defined.
3. CPIR-L4 gives finite-horizon path push-forward commutation for all registered compatible strategies and initial distributions; D7 pushes the equality to target response distributions.
4. The history-response quotient is a space of equivalence classes and, by definition, preserves the responses registered in the task Θ .

5. The inverse images or set-valued images of $q^E, \Phi^H, \tau, \Gamma$ give the lower-layer implementation fibers of events, histories, states, and actions explicitly; Θ gives admissible strategies, targets, environment, distances, horizon, and error budgets, so the effective domains are explicit.

The first four properties belong to different types, and no equality between event poset edges and state transition edges is used in combining them. If an implementation relation χ is additionally claimed for structure–dynamics, its cross-quotient compatibility is an extra premise; the theorem only carries the given certificate and does not derive it from L1–L4. All five properties follow.

T1 is a modular stitching theorem on finite objects. Its content is to prove that four quotient types and maps can coexist within a common task contract while preserving their respective types, not to derive Markov dynamics from posets alone.

B.10.7 CPIR-C1: Persistence Probability Preservation

Corollary CPIR-C1. Under the CPIR-T1 conditions, suppose the persistence indicator satisfies $G_T = \bar{G}_T \circ \tau^{0:T}$. Then

$$P_{\mathcal{E}}^{\mu, \pi}(G_T = 1) = P_{\mathcal{E}}^{\tau_{\#}\mu, \bar{\pi}}(\bar{G}_T = 1).$$

Proof. In CPIR-L4 take $F = G_T$, $\bar{F} = \bar{G}_T$, and the outcome set $A = \{1\}$; the response commutation equation immediately gives the conclusion.

Here “descends to the quotient” means that G_T factors through $\tau^{0:T}$; it does not mean that the persistence probability numerically decreases. If two in-block micro histories have different identity judgments, $G_T = \bar{G}_T \circ \tau^{0:T}$ fails and C1 cannot be invoked.

B.11 The CPIR-T2 Proof Package (conditional DCS result)

B.11.1 The Dobrushin Contraction Lemma (external standard result)

For a Markov kernel K on a finite set Z , define $\alpha(K) = \sup_{z, z' \in Z} d_{TV}(K(z, \cdot), K(z', \cdot))$.

Lemma. For any probability distributions μ, ν : $d_{TV}(\mu K, \nu K) \leq \alpha(K) d_{TV}(\mu, \nu)$.

Proof. Let $r(z) = \min\{\mu(z), \nu(z)\}$ and write $c = \sum_z r(z) = 1 - d_{TV}(\mu, \nu)$. When $c = 1$ the claim is trivial; otherwise write $\mu = c\rho + (1 - c)\mu'$, $\nu = c\rho + (1 - c)\nu'$ with ρ, μ', ν' probability distributions. By convexity, any mixture of two kernel rows has total variation distance at most the kernel row diameter $\alpha(K)$, so $d_{TV}(\mu K, \nu K) = (1 - c)d_{TV}(\mu' K, \nu' K) \leq (1 - c)\alpha(K)$. Substituting $1 - c = d_{TV}(\mu, \nu)$ gives the claim.

B.11.2 Lifting One-Step Defects to Distribution Defects

If $\sup_x d_{\text{TV}}(\tau_{\#}P_t(x, \cdot), \bar{P}_t(\tau x, \cdot)) \leq \delta_t$, then for any micro distribution μ : $d_{\text{TV}}(\tau_{\#}(\mu P_t), (\tau_{\#}\mu)\bar{P}_t) \leq \delta_t$.

Proof. The left-hand side compares convex combinations, with the same weights $\mu(x)$, of paired probability rows. Convexity of total variation gives

$$d_{\text{TV}}\left(\sum_x \mu(x) \tau_{\#}P_t(x, \cdot), \sum_x \mu(x) \bar{P}_t(\tau x, \cdot)\right) \leq \sum_x \mu(x) \delta_t = \delta_t.$$

B.11.3 Proof of CPIR-T2

By the triangle inequality, the distribution-defect lifting lemma, and the contraction lemma:

$$\begin{aligned} e_{t+1} &= d_{\text{TV}}(\tau_{\#}(\mu_t P_t), \bar{\mu}_t \bar{P}_t) \\ &\leq d_{\text{TV}}(\tau_{\#}(\mu_t P_t), (\tau_{\#}\mu_t)\bar{P}_t) + d_{\text{TV}}((\tau_{\#}\mu_t)\bar{P}_t, \bar{\mu}_t \bar{P}_t) \\ &\leq \delta_t + \alpha_t e_t. \end{aligned}$$

Expanding the recursion yields

$$e_T \leq e_0 \prod_{j=0}^{T-1} \alpha_j + \sum_{k=0}^{T-1} \delta_k \prod_{j=k+1}^{T-1} \alpha_j.$$

When $\alpha_t \leq \alpha < 1$ and $\delta_t \leq \delta$, summing the geometric series gives the contractive bound; when $\alpha = 1$, the linear finite-horizon bound follows.

B.11.4 Response and Approximately Measurable Targets

If $f : Z \rightarrow [0, 1]$, the variational form of total variation directly gives $|\mathbb{E}_{\tau_{\#}\mu_T} f - \mathbb{E}_{\bar{\mu}_T} f| \leq e_T$. If the micro target F satisfies $\sup_x |F(x) - f(\tau x)| \leq \varepsilon_F$, then the triangle inequality gives $|\mathbb{E}_{\mu_T} F - \mathbb{E}_{\bar{\mu}_T} f| \leq \varepsilon_F + e_T$. Persistence events are the registered instance of $F \in \{0, 1\}$ in this conclusion. If micro states in the same macro block have different targets or identities, ε_F can reach 1, and dynamical closure cannot substitute for a target measurability certificate.

B.11.5 Source Identity and Machine Regression

The Dobrushin contraction lemma and the telescoping error recursion are standard Markov contraction/perturbation tools; Andrieux gives a published finite-time error bound for coarse-grained hidden Markov dynamics (Andrieux 2012). The DCS contribution of CPIR-T2 is to connect state abstraction, action implementation, response targets, and persistence events to the same type-safe contract. `models/cpir_approximate/` verifies the error bound time step by time step for the finite instance $\alpha = 0.5$, $\delta = 0.02$, and freezes the linear exit at $\alpha = 1$ and the non-measurable-target counterexample. Machine regression only verifies implementation and boundary samples; the F0 identity comes from the finite mathematical proofs above.

This appendix collects all proofs invoked by the main text: C1-T1/T2, C2-P1, C3-T1/T2/T3, and C4-T1 as inherited external standard results; CP-T0, CP-T1, CPIR-L3/L4, CPIR-T1/C1, and CPIR-T2 as conditional DCS results. All F0 labels mean only that conventional proofs are complete and human-reviewed; formalization remains a long-term task.

Appendix C Mandatory Counterexamples and Failure Exits

Each counterexample states: what it negates, what it does not negate, and which map type or contract it requires to be refined.

C.1 C1 Counterexamples

D-C1-01 Local finiteness does not imply finite past. Any interval of $(\mathbb{Z}, <)$ is finite, yet every point has infinitely many past and future points. Prevents replacing finiteness with local finiteness.

D-C1-02 Isospectral non-isomorphic. Take a known isospectral graph pair; even with identical spectra, structures may differ. Regression target: code must not decide isomorphism by spectral hash.

C.2 C2/C3 Counterexamples

D-C2-01 Hasse insertion. Initial cover $a \rightarrow c$; inserting $a \rightarrow b \rightarrow c$ makes the old edge non-covering. Prevents local additions without reduction.

D-C3-01 Shannon entropy decrease. A two-state chain evolving from uniform to a highly skewed stationary distribution can decrease Shannon entropy. Prevents “structural entropy always increases.”

D-C3-02 Growth without stationarity. Monotone node-count growth makes size drift upward; a fixed full-state stationary distribution generally does not exist. Prevents extrapolating F0 theorems to G.

C.3 C4/C5 Counterexamples

D-C4-01 Coarse-graining generates memory. A three-state chain merges two micro states with different departure rates into one block; the observed block sojourn time is non-exponential. Prevents automatic Markov closure.

D-C4-02 No unique scale. Two independent slow processes have different time constants; task A/B selects different macro variables. Prevents a unique endogenous scale.

D-C5-01 Stationary without organization. A fast uniform random walk on a complete graph has a stationary distribution but no local persistent substructure.

D-C5-02 Same topology, different lifetimes. Identical state graphs with different rates yield metastable lifetimes differing by arbitrary orders of magnitude. Prevents topological spectra alone from determining stability.

C.4 E-Layer Counterexamples

D-E1-01 Constant-environment false closure. An invariant environment makes conditional mutual information zero while the system depends entirely on external support. Requires environmental interventions.

D-E2-01 Replication without evolution. All offspring identical with equal net growth: replication exists but no selection.

D-E3-01 Compression without meaning. A constant encoding compresses maximally but has zero future information.

D-E4-01 Prediction harmful. With i.i.d. environment and costly computation, a predictor lowers net payoff.

C.5 RQ/RP and SCG Counterexamples

D-RQ-01 Classical mixtures have no interference. Ordinary probabilistic convex combinations have no phases and cannot produce amplitude cancellation or Bell/contextual structure.

D-RP-01 Gradient curl is zero. If $f \in C^2(U)$ on a singularity-free region U , then $\nabla \times \nabla f = 0$. This identity says any model deriving nonzero vorticity directly from $\nabla \ln \rho$ must declare multi-valued phases, singularities, defect domains, or a connection; writing only “closed loop” is insufficient. This counterexample directly audits the scalar-vorticity narrative of SCG (Hallman 2025a, 2025b).

D-RP-02 An ordinary single-valued scalar does not allow half-integer winding. For a single-valued complex scalar $\psi = Re^{i\theta}$, closed-loop continuity gives $\oint d\theta = 2\pi n$ with $n \in \mathbb{Z}$. Half-integer spin requires spinors, double covers, configuration-space quantization, or equivalent structure.

D-RP-03 Higher-order action counterexample. If a Lagrangian density non-degenerately contains $(\nabla^2 \rho)^2$, the variation typically produces a fourth-order spatial equation. Calling it a “second-order governing equation” requires demonstrating a degeneracy constraint or boundary simplification.

D-RP-04 The SCG galaxy limit. For the SCG candidate profile $\rho = \rho_0 e^{-B/r}$, one has $\partial_r \ln \rho = B/r^2$; setting $v^2/r = c^2 B/r^2$ gives $v = c\sqrt{B/r} \rightarrow 0$, not a large-radius flat rotation curve. Any “automatic explanation of flat curves” claim must replace the profile and fit out-of-sample.

D-RP-05 Fitted constants are not invariants. A ratio that becomes near-constant after reselecting scale, order, or exponent on different data is not automatically a dimensionless universal constant. It requires a prior definition, error bars, independent samples, and comparison statistics. This audits the γ_{cause} candidate (Hallman 2025c).

C.6 RB/ROOL Counterexamples

D-RB-01 A stable flame is not replicative life. Open dissipative structures can persist with boundaries yet have no heritable replication.

D-ROOL-01 Hypercycle parasite collapse. A parasite that replicates fast but does not catalyze its successor can invade an ideal hypercycle; compartments or spatial structure mitigate only under specific migration, splitting, and decay parameters (Szathmary and Demeter 1987; Boerlijst and Hogeweg 1991; Szostak et al. 2016).

D-ROOL-02 Small self-replicating reactions are not autonomous life. QT45 template-dependent self-sequencing/complementary-strand synthesis is direct experimental progress, but has not simultaneously realized self-sufficient substrates, sustained multi-round replication, mutation-selection, compartmental metabolism, and open-ended evolution (Gianni et al. 2026).

D-ROOL-03 Growth-first is not universal. If replication/translation functions are easier to reach than growth, or if protocell division and resource assumptions change, the “growth before information” order of the Palmeira model can change (Nunes Palmeira et al. 2026).

C.7 RC/RS Counterexamples

D-RC-01 Triune brain counterexample. Vertebrate brain regions are not three independently stacked “reptilian–limbic–neocortical” layers; shared development, extensive interconnection, and functional reuse prevent deriving cognitive hierarchies from that narrative (Cesario et al. 2020).

D-RC-02 Criticality false positives. Downsampling, non-stationary driving, and shared inputs can produce approximate power laws; slightly subcritical states also fit in vivo avalanche data (Priesemann et al. 2014).

D-RS-01 Social brain competing explanations. Sociality–brain-size associations cannot exclude diet, foraging, ecological difficulty, body mass, life history, and culture (Dunbar 2024; De-Casien et al. 2017; Gonzalez-Forero and Gardner 2018).

D-RS-02 Network hierarchy is not a natural constant. Relation definitions, platform caps, rounding, sample selection, and maintenance costs can produce apparent nested hierarchies. The Dunbar number can only be an object of distributions and model comparison.

C.8 Ethical Counterexamples

High mutual information in modules is not evidence of consciousness; high-survival institutions can be unjust; collective stability does not imply individual well-being. Any inference from stability to value requires an independent normative premise.

C.9 The C_{PIR}-X1–X8 Counterexample Table

The “killed inference—minimal construction—failing field—repair exit” for each counterexample is split into two tables to keep every cell readable. Table C.1 records the structure that each counterexample defeats; Table C.2 records the failing field and the repair exit.

Table C.1: Inference killed and minimal construction

ID	Inference killed	Minimal construction
X1	arbitrary partitions automatically yield lawful poset quotients	quotient graph has bidirectional edges
X2	certificates remain valid after SCC repair	SCC of two blocks merges into one
X3	arbitrary partitions automatically yield Markov macro kernels	different rates within block
X4	macro actions naturally have unique micro-implementation effects	two implementations of one macro action differ
X5	dynamical closure automatically guarantees task sufficiency	closed but target depends on compressed variables
X6	one-step fits automatically guarantee multi-step memorylessness	one-step equal, two-step unequal
X7	dynamical quotients automatically preserve persistence identity	identity judgments differ within block
X8	descriptive compression automatically generates new capabilities	compression ratio 2.0 but reachable output set unchanged

Table C.2: Failing field and repair exit

ID	Key failing field	Repair exit
X1	macro_cycle: true	require AEQ acyclicity and recertify
X2	requires_recertification: true	rerun all certifications after SCC repair
X3	lumpability_defect: 0.8	require D5 per-implementation strong blockability
X4	implementation_invariant: false	require D6 multi-implementation invariance

ID	Key failing field	Repair exit
X5	target_measurability_defect: 1.0	require D7 target measurability
X6	two_step_equal: false	enter D2 memory augmentation
X7	persistence_measurability_defect: 1.0	require D8 persistence measurability
X8	new_reachable_capability: false	capability baseline and ablation checks

C.10 History Dynamics, Closure, and RG Counterexamples

D-RQ-02 BDG actions are not arbitrary sums of edge phases. BD/BDG actions depend on specific layer/interval counts and dimension correlation coefficients; assigning arbitrary phases to Hasse edges and summing does not automatically reproduce the functional or its continuum limit (Benincasa and Dowker 2010; Machet and Wang 2021).

D-RQ-03 Known suppression is not the full bad-order-poset suppression theorem. Existing results cover specified bilayer, KR, or layered classes and parameter intervals; local class results must not be promoted to “all non-manifold causal sets are excluded” (Loomis and Carlip 2018; Mathur et al. 2021; Carlip et al. 2023, 2024).

D-RQ-04 Closed unitary evolution has no ordinary global attractors. If U_t is unitary, then $\|U_t\psi - U_t\phi\| = \|\psi - \phi\|$. Quantum branches discussing attractors must switch to open systems, decoherence, conditioning, or coarse-grained dynamics.

D-C4-03 A global spectral gap does not guarantee that a specified partition is blockable. Any finite irreducible three-state CTMC can have a positive spectral gap; merging two micro states with different total rates into a target block still violates strong lumpability and produces non-exponential block sojourn times.

D-RG-01 Poisson refinement is not a fixed point without rescaling. Independent p -refinement changes intensity Λ to $p\Lambda$. RG fixed points can only be discussed after registering coordinate/density rescaling and projecting back into the parameterized model family.

D-CSG-01 CSG couplings are not internal graph edge weights. The t_k parameters of classical sequential growth parameterize the whole transition probability law; they live in the law/coupling space, not as weights on cover edges inside a generated causal set (Rideout and Sorkin 2000; Varadarajan and Rideout 2006; Martin et al. 2001).

C.11 Persistence and Causal Emergence Regressions

D-CP-01 High persistence, low EI. Let all $a_i \mapsto a_1$. The candidate pattern never exits, but under uniform intervention the next state carries no input information, so EI is zero.

D-CP-02 Low persistence, high EI. Let $a_i \mapsto b_i \notin A$. The candidate pattern exits in one step, but the next state encodes the input one-to-one, so EI is $\log_2 n$. D-CP-01/02 jointly prevent “high EI equals persistence.”

D-CP-03 Formation rate and lifetime are not identifiable. High-frequency short-lived and low-frequency long-lived patterns can produce identical cross-sectional abundance. Without cohort or formation-process data, stability screening alone cannot identify either.

D-ACTION-01 Stationary action but exponential instability. For the inverted oscillator $L = \frac{1}{2}\dot{q}^2 + \frac{1}{2}\omega^2 q^2$, $q = 0$ satisfies $\delta S = 0$, while perturbations satisfy $\delta\ddot{q} - \omega^2\delta q = 0$, containing exponentially growing solutions.

D-HAM-01 Hamilton stability is not dissipative attraction. Liouville volume conservation prevents global dissipative attractors that swallow positive-volume neighborhoods in the ordinary sense.

D-RT-01 Open but not self-organizing. A driven system remains homogeneous below threshold, showing openness is not a sufficient condition for dissipative structures.

D-RT-02 Strong driving enters chaos. Increasing the control parameter further can drive stationary patterns through secondary bifurcations into defect chaos or turbulence, preventing “stronger driving, more stable.”

D-RT-03 Entropy production extrema decouple from stability. The minimum entropy production theorem holds only under specific linear near-equilibrium contracts; stable branches far from equilibrium can have higher entropy production.

D-RB-02 RAF exists but dynamics go extinct. A reaction graph satisfies the RAF condition, but concentrations decay to zero at sufficiently large dilution, leakage, or parasite rates.

D-RB-03 Negative feedback without self-production. An external device maintaining negative feedback can stabilize variables without generating the system’s own boundary or constraint network.

D-FBA-01 Constraint steady states are not dynamical attractors. Multiple flux vectors satisfy $Sv = 0$; different objective functions select different solutions, and no convergence statement exists without dynamics.

D-CST-01 Arbitrary causal sets need not be manifold-like. Local-finite poset kinematics does not guarantee a faithful manifold embedding, let alone a four-dimensional Lorentz limit.

D-BENTO-01 Past-infinite frameworks do not decide the real universe. Compatible past-infinite covariant growth models exist; they do not imply that the real universe is past-infinite or that the big bang is a causal-set phase transition.

D-CE-01 EI optimization loop. Maximizing EI on training data by algorithm and then citing that same training EI as evidence that natural systems tend toward high EI constitutes a target-evidence loop.

D-CE-02 Good observation prediction but failed intervention commutation. Hidden confounding can give a macro variable excellent prediction while making $C_{\#}P^{do(i)}$ severely inconsistent with $P^{do(\vartheta(i))}$.

C.12 Causal Semantics and History Regressions

D-SEM-01 Correlation is not intervention effect. In $X \leftarrow U \rightarrow Y$, X and Y can be highly correlated while replacing X 's structural equation does not change Y . Prevents claiming real intervention causality directly from edge weights, correlations, Granger, or transfer entropy.

D-HIST-01 Incomparability does not distinguish concurrency from conflict. Events a, b incomparable in the poset may indicate genuine concurrency (both can co-occur) or mutual exclusion (alternatives). Without conflict and enabling relations, a bare poset cannot answer “why the other history did not occur.”

D-VCA-01 Markov equivalence does not allow unique orientation. Different DAGs can share the same skeleton and v-structures. Without interventions, time order, non-Gaussianity, mechanisms, or other additional conditions, discovery algorithms return at most the corresponding equivalence class.

D-CE-03 Exact closure but information poverty. Mapping all micro states to one constant macro state yields a trivial exact Markov closure, but the macro variable has no discrimination, prediction, or control content. Closure is not a sufficient condition for positive causal emergence or higher-layer ontology.

C.13 Frozen Regression Gates

D-CP-01, D-CP-02, D-ACTION-01, D-C4-01, D-RB-02, and D-CST-01 are mandatory regressions before freezing. Any unconditional occurrence in the main text of “EI implies stability,” “stationarity implies attraction,” “RAF implies life,” or “poset implies manifold” must fail the audit. Frozen regressions must also search for and reject: “nonzero TE implies real causality,” “incomparability implies concurrency/spatial separation,” “discovery algorithm output implies identified causal graph,” and “exact closure implies high causal efficacy.”

C.14 Counterexample Admission Process

Each counterexample carries minimal data, expected failure, proposition number, and an automated test. External contributions are admitted by two-person review with contributor credit; affected propositions enter restricted/retired/refuted in the same version, not deferred to the next major revision. Appendix C, the Appendix E proposition ledger, and `registry/propositions.yml` must be updated in the same commit.

This appendix provides the complete counterexample library for C1–C5, the E layers, RQ/RP/SCG, RB/ROOL, RC/RS, history/closure/RG, persistence/causal emergence, and causal semantics, plus the “killed inference—minimal construction—failing field—repair exit” table of CPIR-X1–X8. Each counterexample states what it negates and what it does not negate.

Appendix D Numerical Models, Parameters, Algorithms, and Reproduction

This appendix gives, model by model, the state space, transition kernels, partitions, targets, distances, thresholds, seeds, run commands, output files, positive/counterexample criteria, and result identity. Code need not be printed in full in the PDF, but must give repository-relative paths, version/hash, and minimal reproduction commands.

D.1 The CPIR Finite Exact Quotient Regression Model

Repository path: `models/cpir_exact/`; model ID: CPIR-F0-EXACT-REGRESSION v1.0; type F0; profile CPIR-C; closure class D0-exact.

Purpose. Implements the CPIR-X1–X8 counterexamples and one finite exact positive example registered in 09E and Appendix B. The model uses only finite sets, deterministic maps, and finite Markov kernels, reusing the standard criteria of poset quotients, strong blockability, target measurability, and implementation invariance.

Inferences killed. X1 arbitrary partitions automatically yield lawful poset quotients; X2 certificates remain valid after SCC repair; X3 arbitrary partitions automatically yield Markov macro kernels; X4 macro actions naturally have unique micro-implementation effects; X5 dynamical closure automatically guarantees task sufficiency; X6 one-step fits automatically guarantee multi-step memorylessness; X7 dynamical quotients automatically preserve persistence identity; X8 descriptive compression automatically generates new capabilities.

Result identity. Passing these 12 directed tests only unlocks the machine regression gate for CPIR-T1/C1; theorem identity still requires joint confirmation by the main-text proof, source mapping, and human review.

Run commands.

```
python models/cpir_exact/cpir_exact.py
python -m unittest tests/test_cpir_exact.py
```

What is not established. Continuous or infinite CPIR, quantum CPIR, approximate error bounds, empirical capability emergence.

D.2 The CPIR-T2 Finite Approximate Error Model

Repository path: `models/cpir_approximate/`; model ID: CPIR-APPROX-FINITE-01; type: finite discrete-time classical Markov; distance: total variation; contraction coefficient: Dobrushin.

Registered instance. Four micro states, two macro states; $\alpha = 0.5$, $\delta = 0.02$. The program checks time step by time step that the actual TV error does not exceed the geometric bound, and that macro-measurable targets inherit the bound. Two boundary samples are included: with $\alpha = 1$ only the linear finite-horizon bound is available (0.16 at $T = 8$); and with zero macro distribution error, a hidden, non-macro-measurable target can still reach maximal error (lesson: distribution closure does not certify a non-macro-measurable target).

Representative result table.

Horizon T	Actual error	Theoretical bound
1	0.02	0.02
4	0.0025	0.0375
8	0.000156	0.039844
12	0.000010	0.039990

Run commands.

```
python models/cpir_approximate/cpir_approximate.py
python -m unittest tests/test_cpir_approximate.py
```

Result identity. This artifact verifies formula implementation and counterexample responsibility; it does not constitute empirical confirmation in natural systems; the F0 identity comes from the finite mathematical proof in Appendix B.11.

D.3 Selective Causal Compression Cross-Model Validation

Repository path: `models/selective_compression_validation/`; model IDs: SCC-F1-001 (six-state metastable basins), SCC-RG-F0-001 (eight-state hierarchical bit chains), SCC-F2-001 (eight-state non-product non-reversible circulation chains).

Purpose. Verifies the contract logic of 09D: exact alignment and causal plateaus can appear in controlled families, and provides failure exits after breaking symmetry, product structure, and reversibility.

Pre-registered thresholds. Response distortion at most 0.01; closure defect at most 0.005; two-step memory residual at most 0.005; dynamics-response direction cosine at least 0.99. Registration tolerance 10^{-12} .

Non-reversible circulation scan (heterogeneity h).

h	Response distortion	Closure defect	Memory residual	Direction cosine
0.0	0.0	0.0	0.0	1.0
0.1	0.00860	0.00400	0.00323	0.99998
0.2	0.01730	0.00800	0.00649	0.99993
0.8	0.07415	0.03200	0.02633	0.99889

Joint admission plateau endpoint: $h = 0.1$; from $h = 0.2$ all three errors cross thresholds. **Allowed conclusion:** high directional alignment can coexist with joint interface admission failure; **forbidden conclusion:** a causal plateau in a real natural system has been empirically discovered.

Run command.

```
python models/selective_compression_validation/run_model.py
```

Result identity. Certificates within finite models; they do not represent general complex systems or real physical systems.

D.4 The Cross-Scale Interface Joint Admission Prototype

Repository path: `models/cross_scale_interface/`; model ID: CSR-F0-001; state space $\{A0, A1, B0, B1, D\}$ (with D the absorbing failure state); action set $\{\text{baseline}, \text{protect_a}, \text{protect_b}\}$; future responses are scored on A/B/D outcomes at horizons 1, 2, 3; registered partition $\{A0, A1\}|\{B0, B1\}|\{D\}$; candidate space is all 15 set partitions of the four alive states (D fixed singleton).

Selection rule. Two-stage: first require non-trivial inter-class response, genuine compression, and response, closure, memory, intervention, and persistence errors to pass hard gates (10^{-10}), then compare seven-objective Pareto among admissible partitions. The raw Pareto frontier is fully preserved, to exhibit that “one-block collapse” may remain on the frontier because it minimizes interface cost.

Frozen parameters. `internal_probability` 0.24; `cross_probability` 0.18; `death_A` 0.06; `death_B` 0.10; `protection_death_multiplier` 0.35; `protection_cross_multiplier` 0.80. Perturbation: `signed_internal_index_c` baseline 0.0, counterexample value 0.25.

Results. In the symmetric model the registered partition simultaneously has zero within-class response distortion, zero closure defect, zero two-step memory residual, zero intervention commutation error, and zero persistence distortion (CSR-P1); at within-block heterogeneity 0.25, response, closure, memory, and persistence distortions turn positive while intervention commutation remains zero (CSR-P2), showing that the error axes are independent.

Run command.

```
python models/cross_scale_interface/run_model.py
```

Result identity. The model only establishes finite-state counterexamples and certification baselines; cross-model universality, continuous manifolds, RG fixed points, and life/intelligence explanations remain the burden of subsequent bridging.

D.5 Evidence Ledger Excerpt

`registry/evidence-ledger.yml` (schema v2, updated 2026-09-02) registers all proofs, planned experiments, and literature support. Completed items directly relevant to this preprint: CPIR-L1/L2 are imported specializations; CPIR-L3/L4, CPIR-T1, and CPIR-C1 are human-checked proofs with machine regression (F0); CPIR-T2 is a finite proof with executable regression (F0, 9/9 directed and 105/105 full tests); CPIR-X1–X8 are executable finite counterexample regressions (N1, 12/12 directed and 96/96 full tests); CPIR-H1 is a planned experiment (N0). **verification:** `human_checked` means the conventional proof has been human-reviewed; only `proof_assistant_checked` means machine verification is complete; the two must not be conflated.

This appendix gives complete registration cards, frozen parameters, pre-registered thresholds, representative results, run commands, and result identities for the four model families. All numerical values are certificates within finite models, not empirical confirmation in natural systems; the authoritative sources of code and data are the `models/` and `registry/` directories.

Appendix E Proposition Ledger, Bridging Obligations, and License Governance

E.1 Two-Dimensional Proposition Grading and Passports

Propositions are graded in a two-dimensional grid of “evidence strength $\times$ theory layer”: the horizontal axis is E0 conjecture, E1 conditional proof, E2 strict theorem, E3 empirical verification; the vertical axis is T1 foundational structure, T2 core causality, T3 organization, T4 cross-domain, T5 about the theory itself. Every formal proposition has a stable ID, status, owner, summary, full statement, supporting evidence, counterexamples, dependencies, and legacy debt; status transitions require a transfer checklist and a new passport. The registry `registry/propositions.yml` is the sole authority.

Identity adjudication rule. When the main text, appendices, and registry conflict, the adjudication of `claim-audit.yml` outranks `propositions.yml` and the book chapter text; this preprint uses `claim-audit.yml` as the sole identity authority. The abstract must not change the identity of frozen propositions.

E.2 Evidence Level and Claim Strength Correspondence

Evidence level	Semantics	Allowed main-text phrasing
E0	conjecture	“we conjecture” “planned verification”
E1	conditional proof	“proved under premise P”
E2	strict theorem	“theorem” (limited to finite, registered scope)
E3	empirical verification	data and statistics required
external standard result	mature tool inheritance	source explicitly marked

E.3 Key Proposition List (Scope of This Preprint)

Proposition	Level	Status	Status note
C1-T1 linear extension	external standard result	active	finite posets
C1-T2 unique transitive reduction	external standard result	active	finite DAGs

Proposition	Level	Status	Status note
C2-P1 lawful rule closure	E1	active	conditional DCS
C3-T1 conservative Q stochastic semigroup	external standard result	active	finite conservative
C3-T2 irreducible stationarity and convergence	external standard result	active	finite irreducible
C3-T3 KL contraction	external standard result	active	Markov data processing
C4-T1 strong lumpability	external standard result	active	fixed partition
C4I-T1 abstraction composition preservation	E1	active	two-level exact, compatible
CP-T0 survivor conditioning enrichment	E1	active	conditional DCS
CP-T1 persistence–EI four quadrants	E1	active	finitely realizable
CPIR-L1 AEQ preserves posets	E1	active	finite quotients
CPIR-L3 macro kernels well defined	E1	active	D5/D6
CPIR-L4 finite-horizon commutation	E1	active	compatible strategy–implementation
CPIR-T1 exact interface quotient	E1	active	six premises, conditional DCS
CPIR-C1 persistence probability preservation	E1	active	quotient-measurable
CPIR-T2 finite approximate error bound	E1	active	finite Markov, TV
CPIR-X1–X8 counterexamples	E1/N1	active	machine regression passed
SCC-H1/H2/H3 conditional plateaus	E0	conjecture	conjecture, not theorem
CE-H1 EI positive association	E0	conjecture	conjecture
P2B finite interconnection prototype	—	future_work	not completed
life dual contract	—	future_work	not completed
intelligence capability lattice	—	future_work	not completed

Proposition	Level	Status	Status note
5.6 physics bridging (RQ/RST/RP)	—	future_work	not completed
6.0 discriminating empirical predictions	—	future_work	does not exist

E.4 Persistence Passport Specification

The **PersistenceClaim** registers: object (pattern/structure/organization); identity relation (the relation deciding sameness); model (F0/F1/G/O/A); scale (time window, spatial domain); environment (perturbations, parameters, noise); time window; perturbation type and magnitude; indicator ($S_A(T)$, survival function, recovery time); cost (C_{maint}); null model; evidence type and association; failure conditions. Defects: no null model registered; substituting LTR, output ELBO, or high S_A directly for persistence; merging the persistence passport with the **E/Q/K** vectors into one “total score.”

E.5 Bridging Obligations, Analogy Licenses, and Counterexample Writeback

Cross-domain bridging papers must fill in: source identity (source and target models), the original proposition number of the borrowed conclusion, the range of universal quantification, empirical or definitional defects, topological/structural correspondence, explanation direction, bridging map scope, string/term/existence conflicts, counterexample writeback entries, author numbers, new-task-new-data declarations, and update dates. An **analogy license** authorizes analogy only; all cross-domain causal claims must separately cite experimental evidence in each domain.

E.6 Explanation Directions and Forbidden Phrasing

Explanation directions include: grounding-from-base explanations, top-down constraints, bidirectional satisfaction, joint optimization, formal analogy, counterfactual correspondence, and ontological construction; without a registered explanation direction, one must not claim “explanation.” Forbidden phrasing: writing “the theory is verified by data” for “the model is fitted to data”; writing “structurally open” as “the system is intelligent”; “tolerance-free” only when determined by discrete steps; “emergence” requires a macro variable, a coarse description, and a registered target appearing outside the source dynamics. Any phrasing reaching “scientific consensus,” “unified explanation achieved,” or “the theory proves the essence of the universe or consciousness” immediately triggers the downgrade process.

This appendix gives two-dimensional proposition grading and passport specification, evidence-level correspondence, the proposition list and status within the scope of this preprint, the persistence passport fields, bridging obligation/analogy license/counterexample writeback governance, and the explanation direction with forbidden phrasing list. `claim-audit.yml` is the identity authority; `propositions.yml` is the sole proposition registry.

Appendix F Glossary, Writing Rules, and Extended Reading

F.1 Chinese–English Term Correspondence Table (DCS 5.5 Core)

English	Chinese	Note
Dynamical Causal Structuralism	因果动力学结构论	DCS
structural reachability causality	结构—可达因果	poset layer
generative history dynamics	生成—历史动力	rewrite/history layer
response-intervention causality	响应—干预因果	action layer
event quotient	事件商	q_r^E
history-response quotient	历史响应商	Φ_r^H
state abstraction	状态抽象	τ_r^X
intervention implementation	干预实现	Γ_r^A
task contract	任务合同	Θ_r/Θ
frozen task contract	冻结任务合同	unchanged during release
conditional causal persistence	条件性因果持存	CP
persistence passport	持存护照	PersistenceClaim
survivor conditioning enrichment	幸存者条件富集	CP-T0
strong lumpability	强可块化	D5
multi-implementation invariance	多实现不变性	D6
target measurability	目标可测性	D7
persistence measurability	持存可测性	D8
finite exact interface quotient	有限精确接口商	CPIR-T1
finite approximate error bound	有限近似误差界	CPIR-T2
total variation distance	总变差距离	d_{TV}
Dobrushin contraction coefficient	Dobrushin 收缩系数	α
admissible event quotient	允许事件商	AEQ
interface certification	接口认证	Cert_Θ
causal role promotion	因果角色上移	ι_r
memory augmentation	记忆增广	D2
analogy license	类比许可证	bridging governance
counterexample writeback	反例回写	governance loop

F.2 Term Usage Rules

- The same term keeps the same translation within the same chapter; the English original and abbreviation are given at first appearance.

- “Emergence” requires a macro variable, a coarse description, and a registered target appearing outside the source dynamics.
- “Fixed point” must distinguish probability-distribution fixed points, generative-process fixed points, and RG semantics.
- “Stable” must specify whether Lyapunov, asymptotic, structural, metastable, or statistical.
- “Attractor” is forbidden without qualification under unitary quantum evolution.
- “Self-organization” and “open” must be registered separately; openness does not imply self-organization.
- Without a registered explanation direction, one must not claim “explanation.”

F.3 Writing Rules Checklist

- Heading levels do not exceed three; the main text is readable without the appendices.
- Every formal result is followed by its “conditions” and “what cannot be concluded.”
- Citation format is uniform; every in-text citation resolves in the reference list (all cite keys resolve; see Appendix E and `references/preprint-v1.bib`).
- Unconditional analogies such as “EI implies stability,” “stationarity implies attraction,” “RAF implies life,” or “poset implies manifold” are forbidden.
- Numbers and facts carry sources; theoretical/model results and empirical results are presented in separate columns.
- The Chinese and English versions have strictly identical claim content; translation must not change any proposition identity.

F.4 Extended Reading

- Causal sets and posets: the Bombelli–Lee–Meyer–Sorkin original and later reviews (Bombelli et al. 1987; Rideout and Sorkin 2000; Benincasa and Dowker 2010; Machet and Wang 2021); causal set suppression literature (Loomis and Carlip 2018; Mathur et al. 2021; Carlip et al. 2023, 2024); classical sequential growth (Rideout and Sorkin 2000; Varadarajan and Rideout 2006; Martin et al. 2001).
- Finite Markov dynamics: Norris, Kemeny–Snell, and strong lumpability (Norris 1997; Kemeny and Snell 1976; Buchholz 1994); coarse-grained hidden Markov error bounds (Andrieux 2012); metastability (Bovier and Hollander 2015).
- Causal abstraction and interventions: Pearl, Rubenstein et al., and Beckers–Eberhardt–Halpern (Pearl 2009; Rubenstein et al. 2017; Beckers and Halpern 2019; Beckers et al. 2020).
- Computational mechanics and predictive compression: Crutchfield computational mechanics, Shalizi–Moore causal states, and the Tishby information bottleneck (Crutchfield and Young 1989; Shalizi and Crutchfield 2001; Tishby et al. 1999).
- Coarse-graining and memory: Zwanzig and Mori projection operators (per the 1972 correction) (Zwanzig 1961; Zwanzig et al. 1972; Mori 1965).
- Causal emergence: the Hoel macro-causal measure (Hoel et al. 2013); its limitations and identification problems appear in Section 8.2 and Appendix C.11.

- Biochemistry and the origin of life: hypercycles and parasites (Szathmary and Demeter 1987; Boerlijst and Hogeweg 1991; Szostak et al. 2016); QT45 template self-synthesis (Gianni et al. 2026); the Palmeira growth-first model (Nunes Palmeira et al. 2026).
- Brain and society: the triune brain critique (Cesario et al. 2020); criticality false positives (Priesemann et al. 2014); the social brain and cognitive ecology (Dunbar 2024; DeCasien et al. 2017; Gonzalez-Forero and Gardner 2018).
- SCG and vorticity formalization: for analysis and counterexamples only; not an endorsement by DCS (Hallman 2025a, 2025b, 2025c).

Appendix F provides the DCS 5.5 core Chinese–English term correspondence, term and writing rules, citation and counterexample obligations, and thematically organized extended reading. Every entry resolves in the reference list; the Chinese and English versions share the same term table.

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